



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3310

Digital Fundamentals and System Design

Wenchao Miao

Outline Lecture 12 Inside logic gates



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1 CMOS (NMOS and PMOS) Transistors

2 CMOS Transistors in Logic Gates

3 CMOS Transistors in Logic Functions

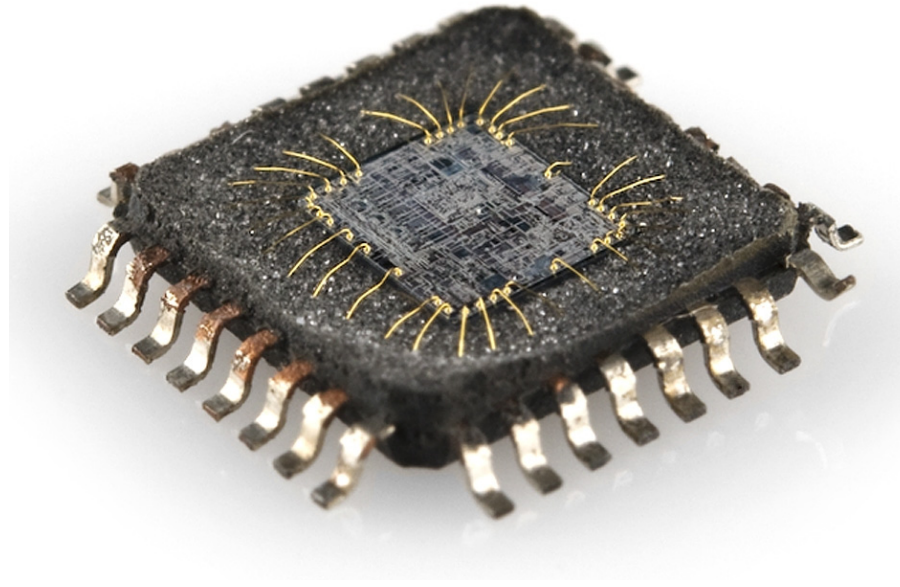
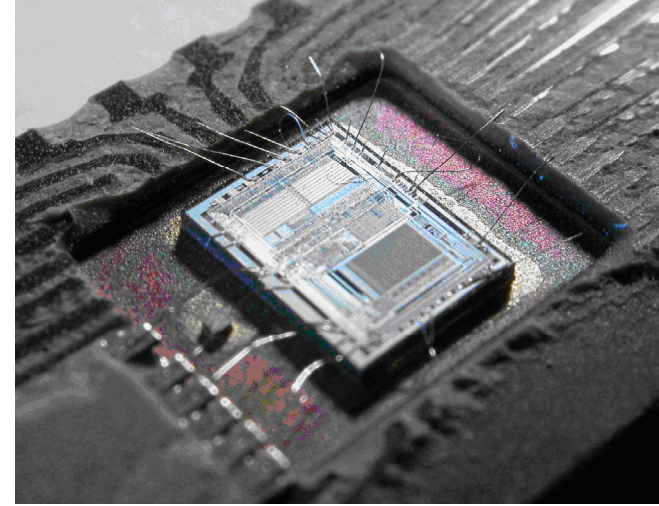
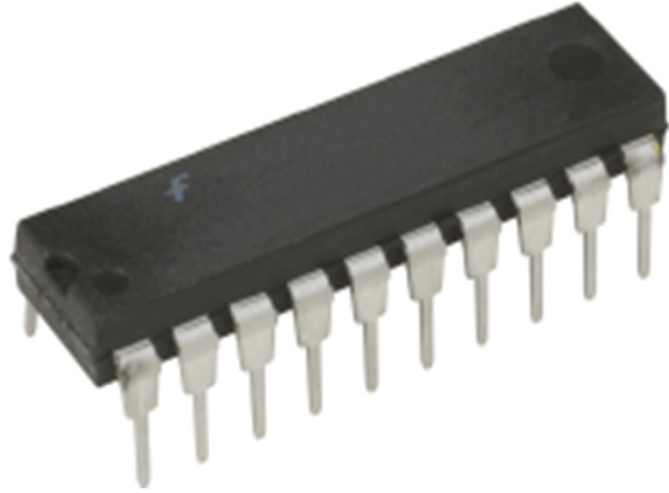
4 Transmission Gates and Tri-state Gates

5 Multiplexers and Demultiplexers

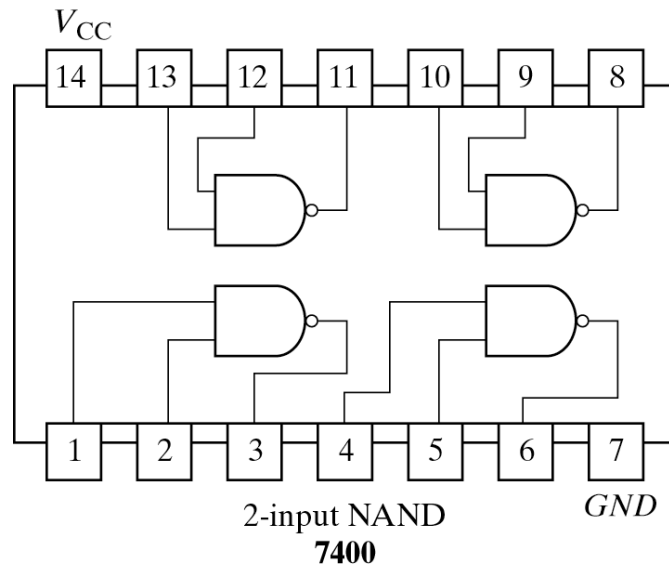
6 Noise Margin and Fan Out

- 1. Recognize CMOS (NMOS and PMOS) transistors**
- 2. Use CMOS transistors to build logic gates**
- 3. Use CMOS transistors to build logic functions**
- 4. Explain the transmission gates and tri-state gates**
- 5. Build multiplexers and demultiplexers using transmission gates and tri-state gates**
- 6. Explain noise margin and fan out**

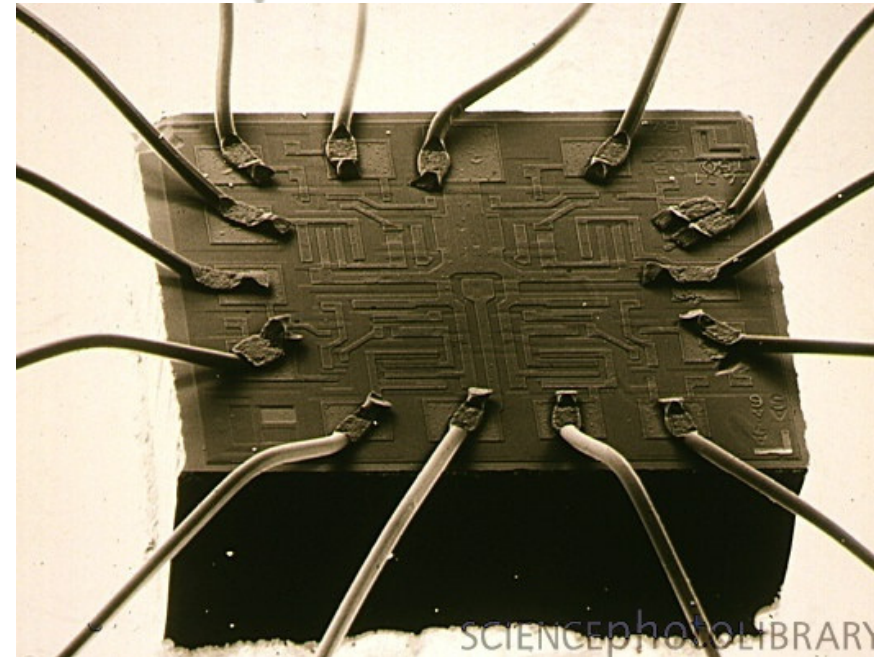
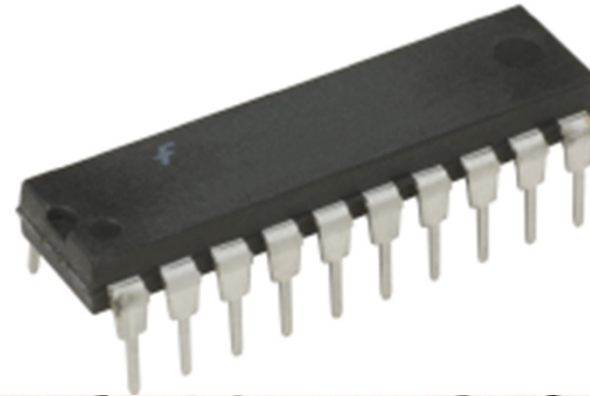
Inside an IC package



Inside 7401 TTL



**7401 TTL integrated circuit (IC)
pictured under electron microscope
→**



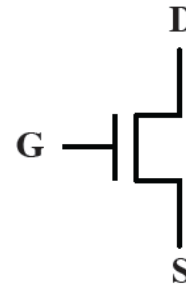
- Transistor as a switch →



- Two types of CMOS transistors
 - PMOS and NMOS transistors complement of each other
 - 3-terminals transistor: Gate (G), Source (S) and Drain (D)

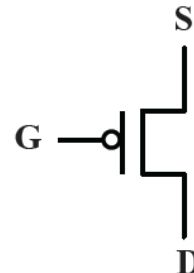
- NMOS transistor

- Circuit Symbol

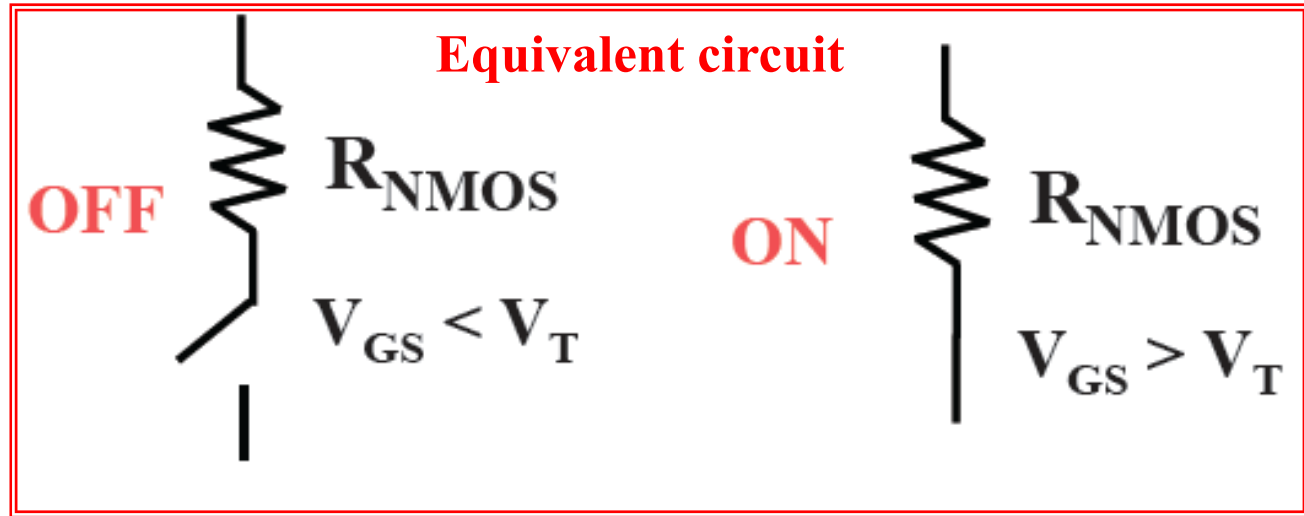
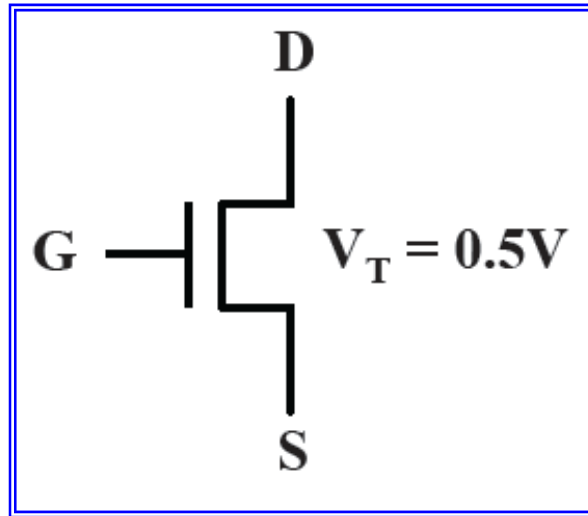


- PMOS transistor

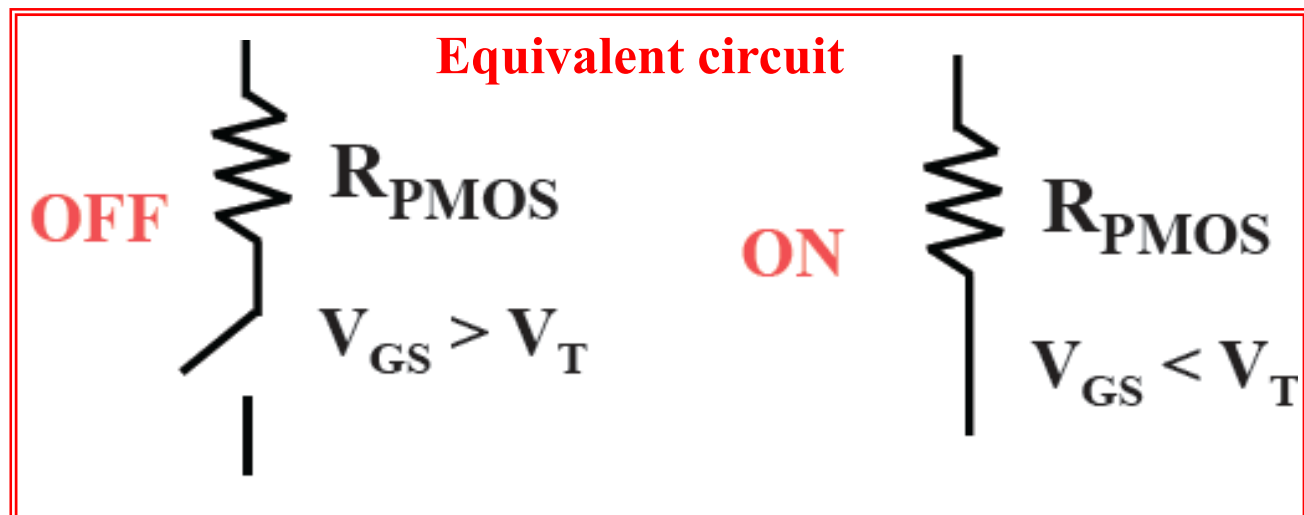
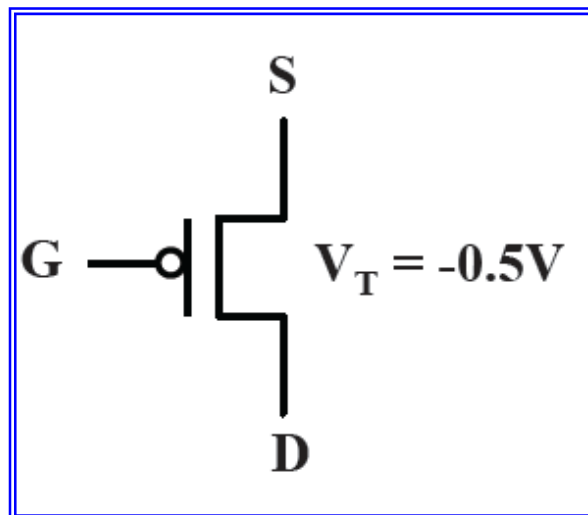
- Circuit Symbol



NMOS Transistor (Switch) is ON when input G is HIGH

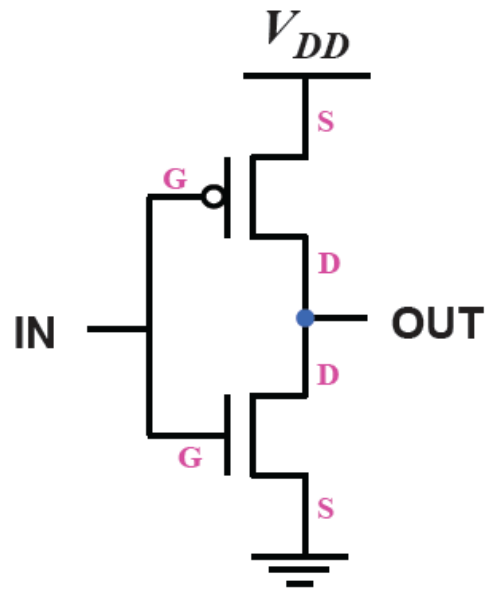


PMOS Transistor (Switch) is ON when input G is LOW



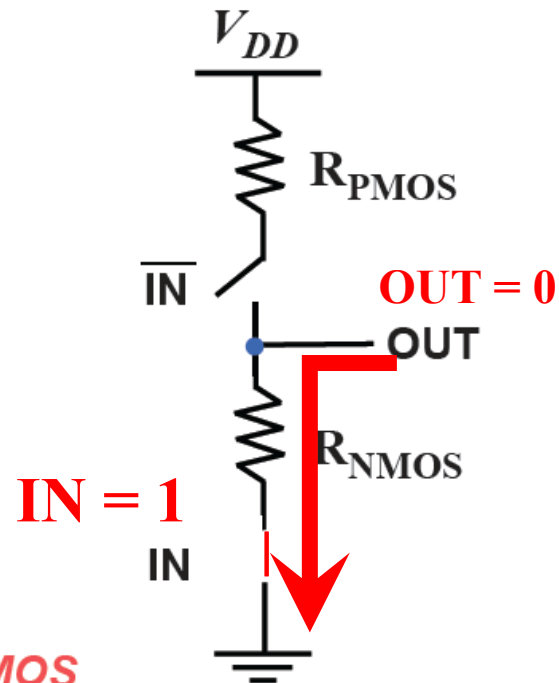
The CMOS Inverter

- When $IN = 1$, NMOS switch is ON (PMOS OFF) $\rightarrow$ output is pulled DOWN to ground (Low) level (Out = 0)
- When $IN = 0$, PMOS switch is ON (NMOS OFF) $\rightarrow$ output is pulled UP to Supply (V_{DD} : High) level (Out = 1)

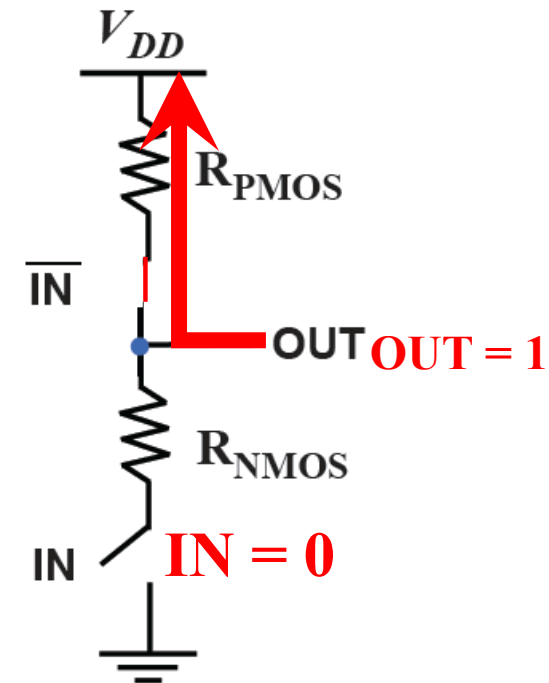


Rail-to-rail Swing in CMOS

Switch Model

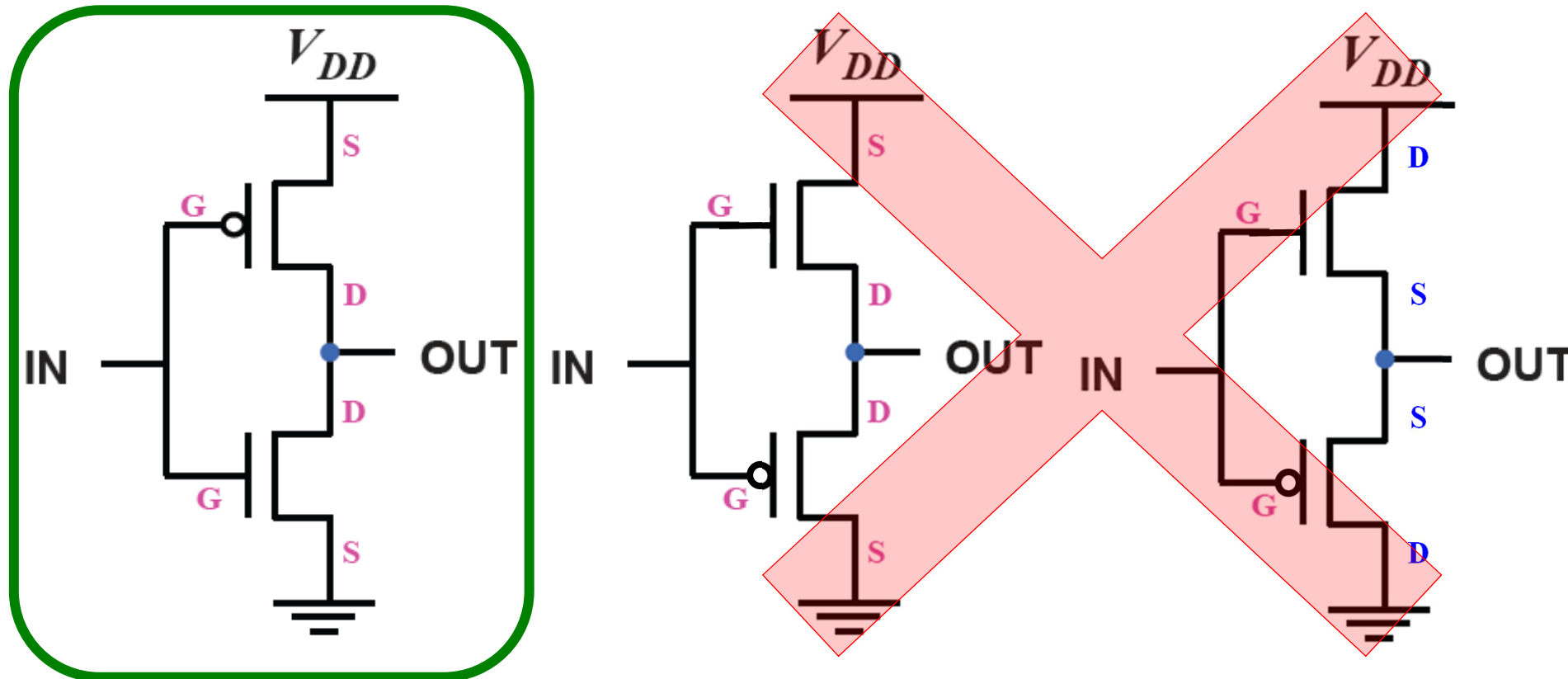


Switch Model



The CMOS Inverter

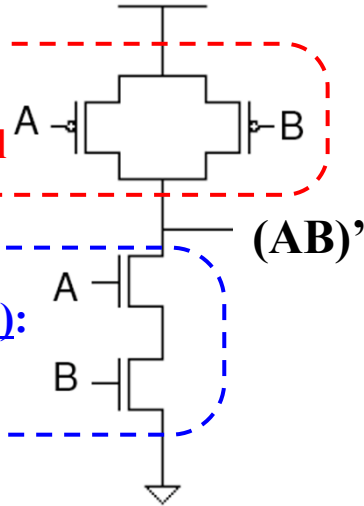
- Always connect PMOS (source input) to V_{DD}
- Always connect NMOS (source input) to Ground
- You cannot swap them. Otherwise, the circuit won't work!



Generic CMOS recipe

NAND

Pull-Up Network (PUN):
PMOS switches in parallel

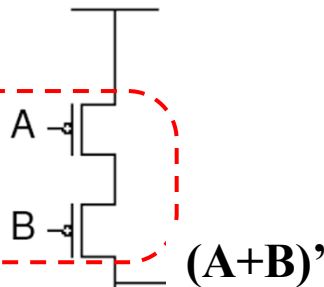


Pull-Down Network (PDN):
NMOS switches in series

A	B	PDN	PUN	Out
0	0	Off	On	1
0	1	Off	On	1
1	0	Off	On	1
1	1	On	Off	0

NOR

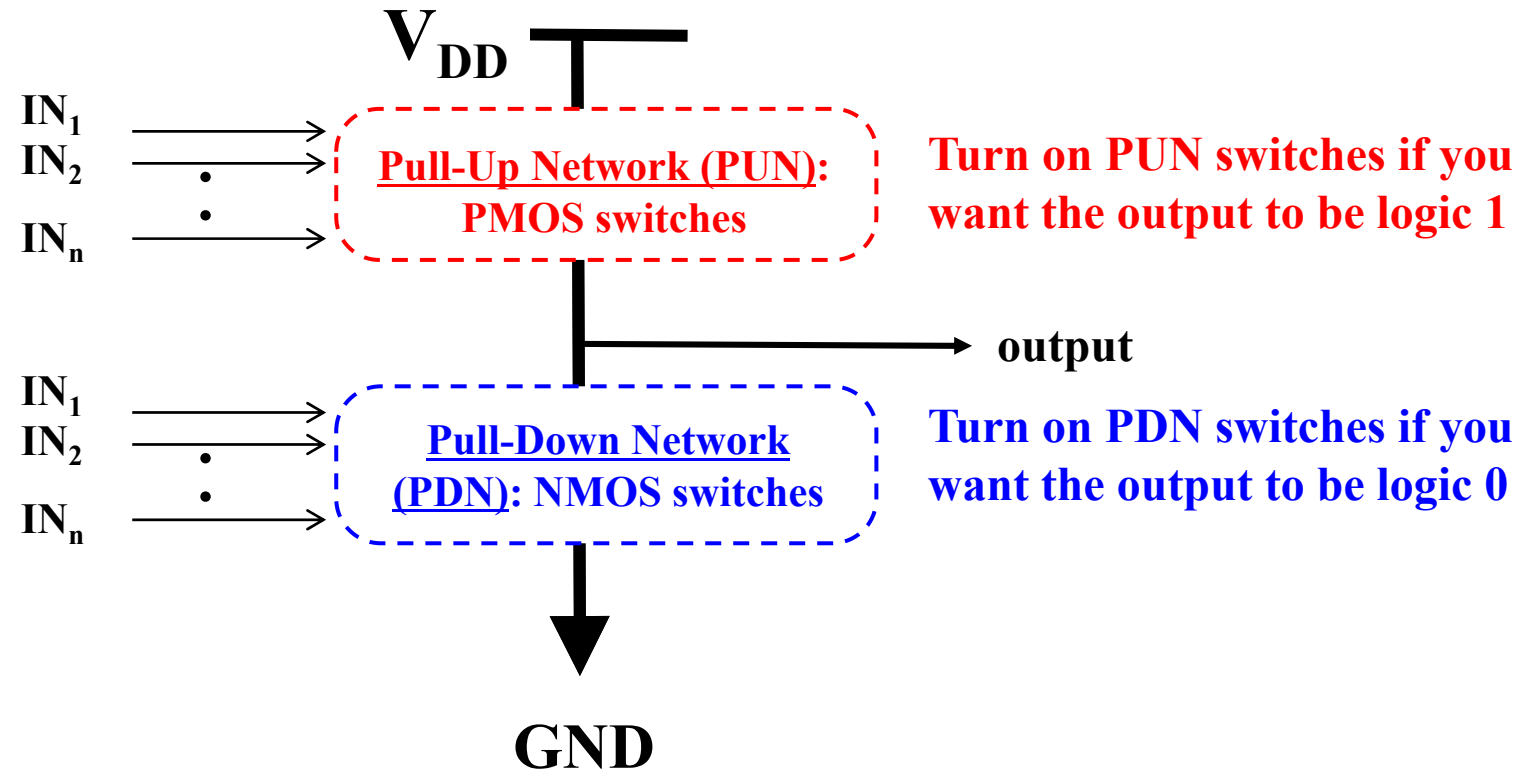
Pull-Up Network (PUN):
PMOS switches in series



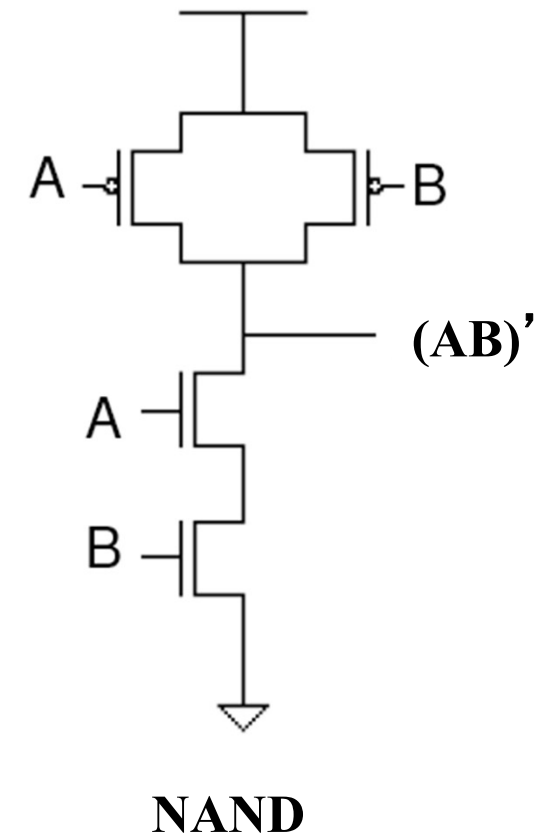
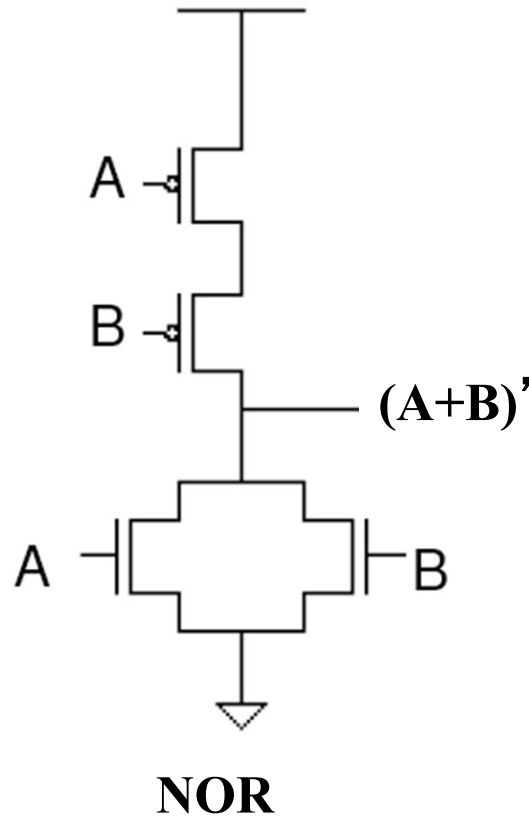
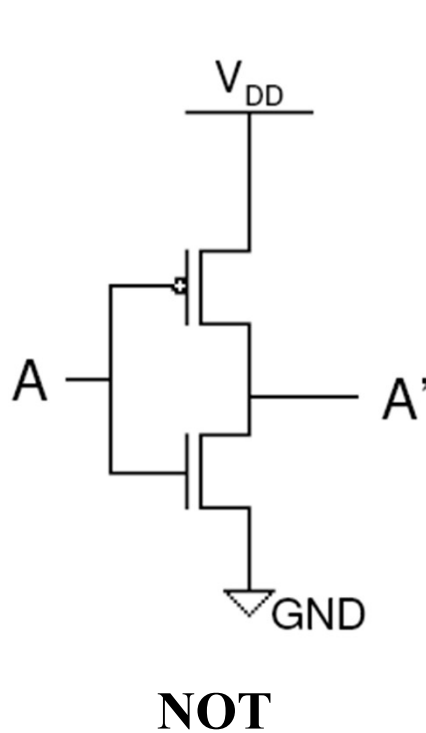
Pull-Down Network (PDN):
NMOS switches in parallel

A	B	PDN	PUN	Out
0	0	Off	On	1
0	1	On	Off	0
1	0	On	Off	0
1	1	On	Off	0

- Make sure PUN and PDN are complementary so that they cannot be ON at the same time
 - If PMOS switches in series $\rightarrow$ NMOS switches should be in parallel
 - If PMOS switches in parallel $\rightarrow$ NMOS switches should be in series



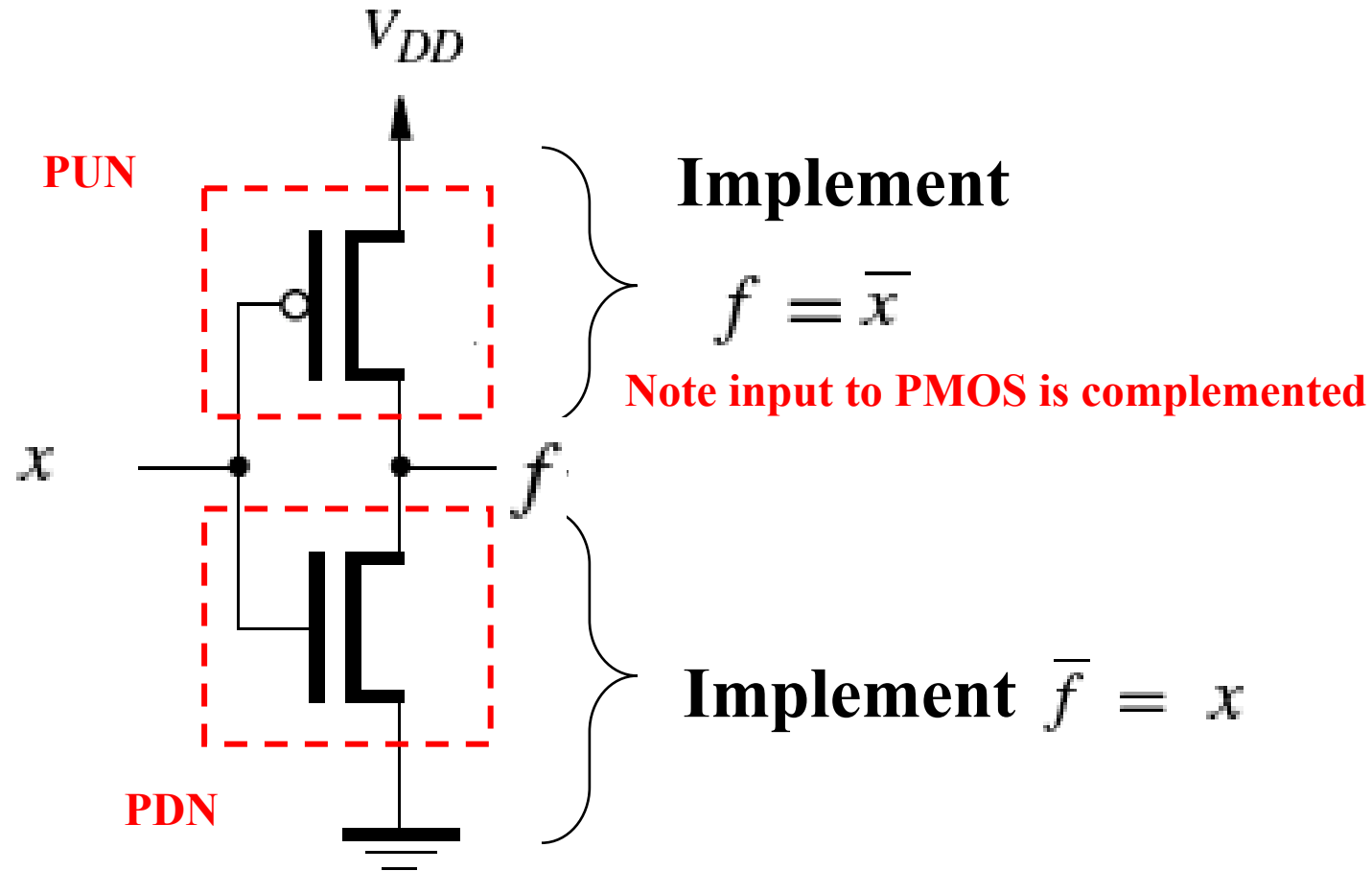
- **NOT, NAND and NOR gates**
 - Notice the outputs are inverted in all CMOS building blocks
 - A 2 input NAND/NOR gate (equivalent gate) requires 4 transistors (switches)



CMOS Realization of NOT (INVERTER)



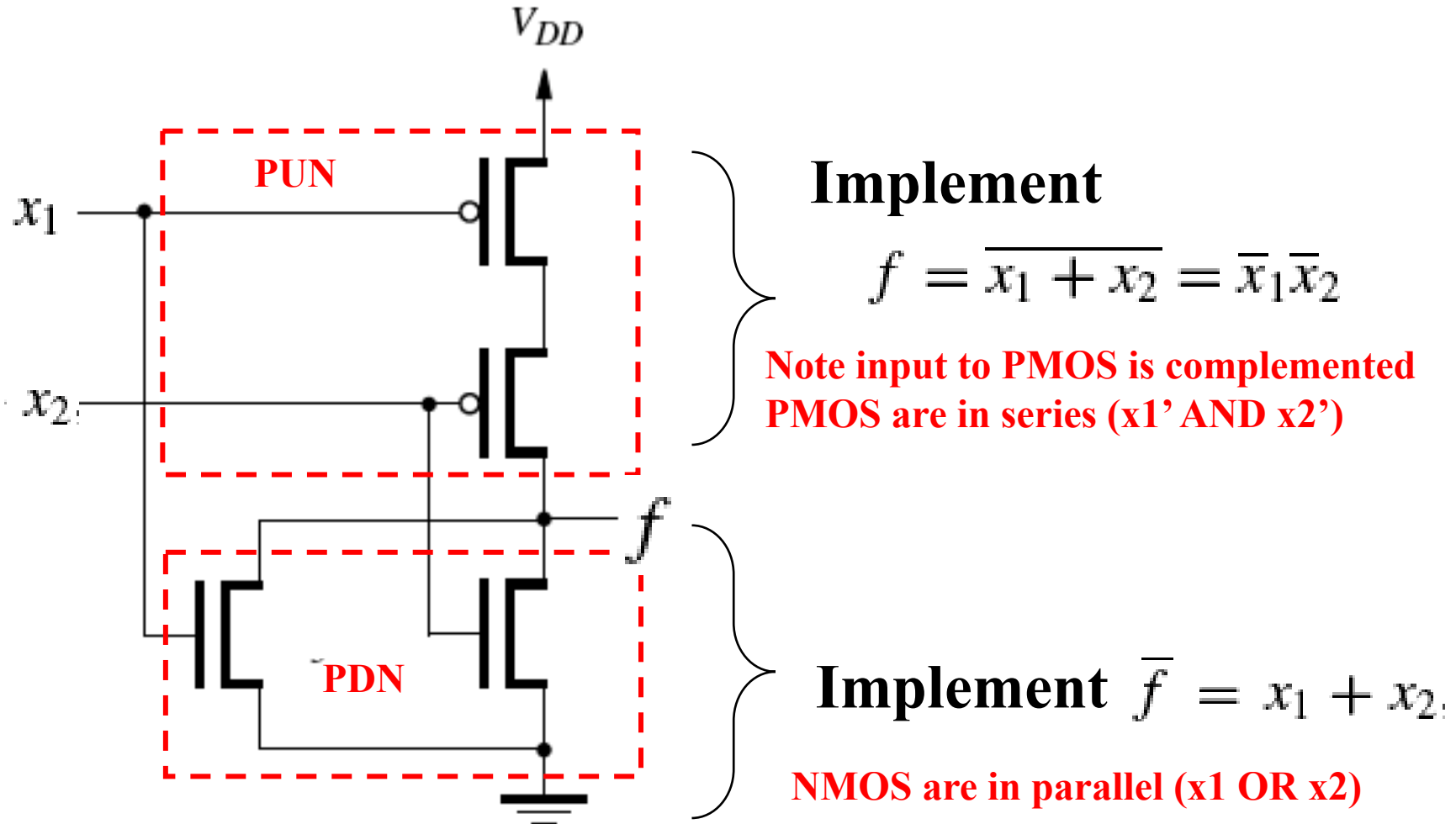
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CMOS Realization of NOR

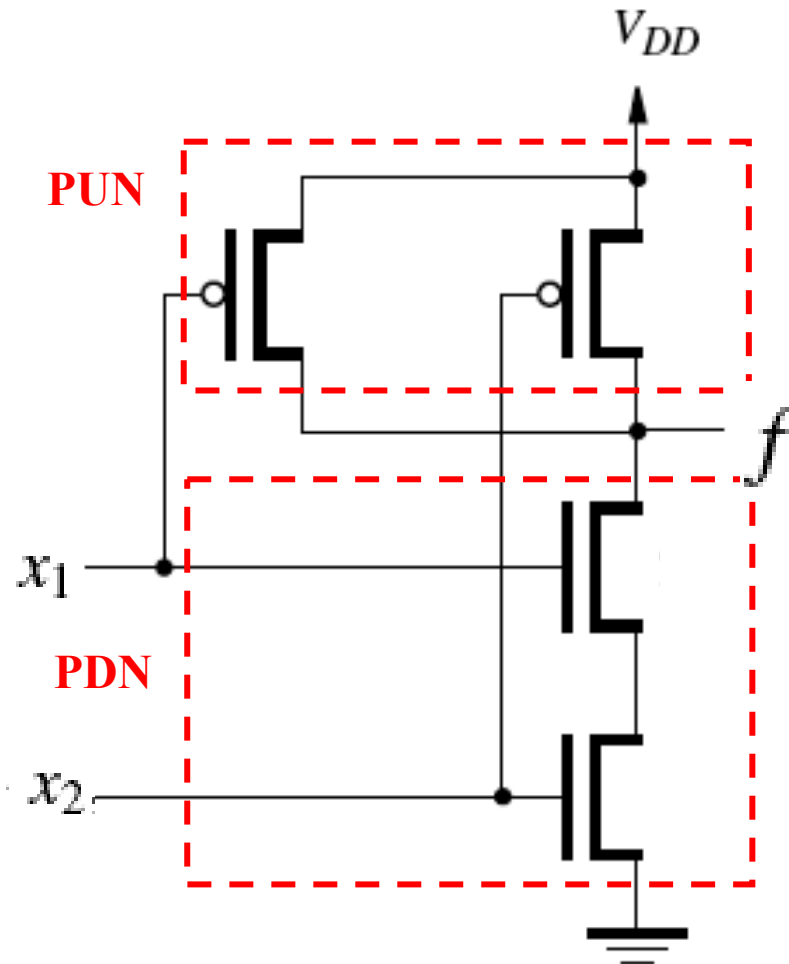
AND operation requires transistors **in series**

OR operation requires transistors **in parallel**



AND operation requires transistors **in series**

OR operation requires transistors **in parallel**



Implement $f = \overline{x_1 x_2}$

$$f = \overline{x_1 x_2} = \overline{x_1} + \overline{x_2}$$

Note input to PMOS is complemented.
PMOS are in parallel (x_1' OR x_2')

Implement $\overline{f} = x_1 x_2$

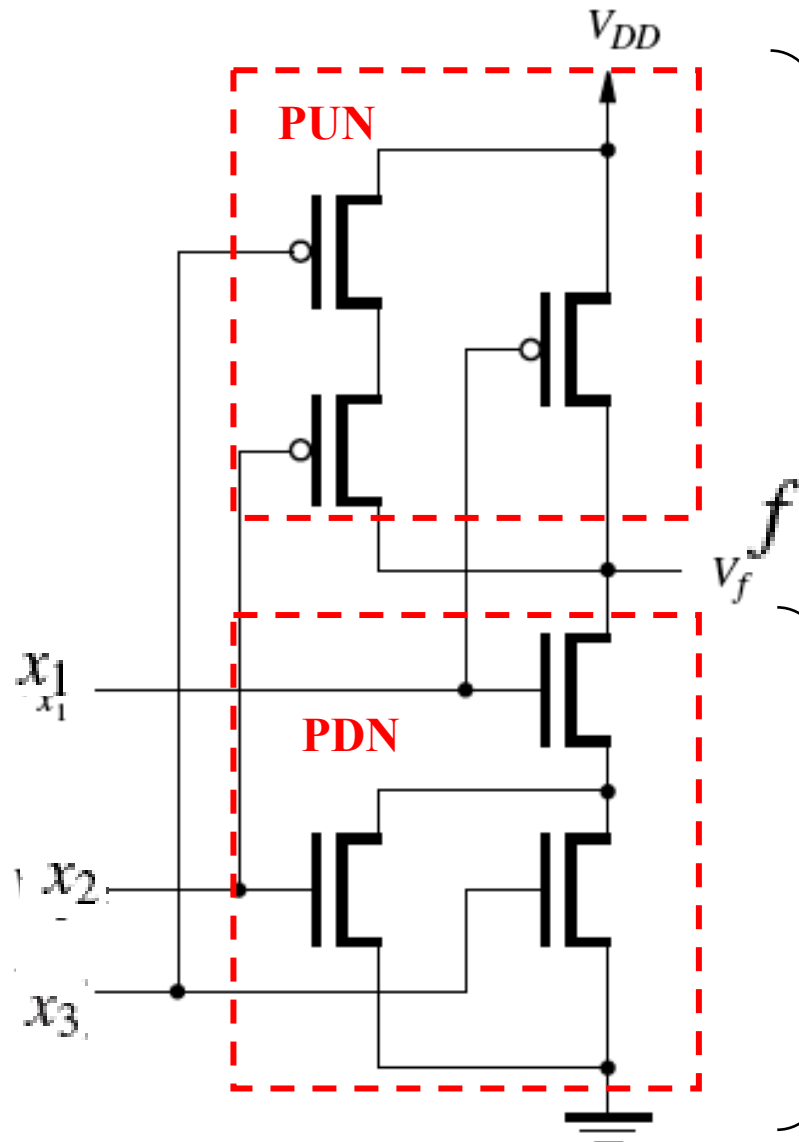
NMOS are in series (x_1 AND x_2)

Example 1

Implement the below function using CMOS logic

$$f = \bar{x}_1 + \bar{x}_2\bar{x}_3$$

Example 1



Implement $f = \bar{x}_1 + \bar{x}_2\bar{x}_3$

All the inputs are already in complement
 x_1' parallel with (x_2' and x_3' in series)

Implement

$$\bar{f} = \overline{\bar{x}_1 + \bar{x}_2\bar{x}_3} = x_1(x_2 + x_3)$$

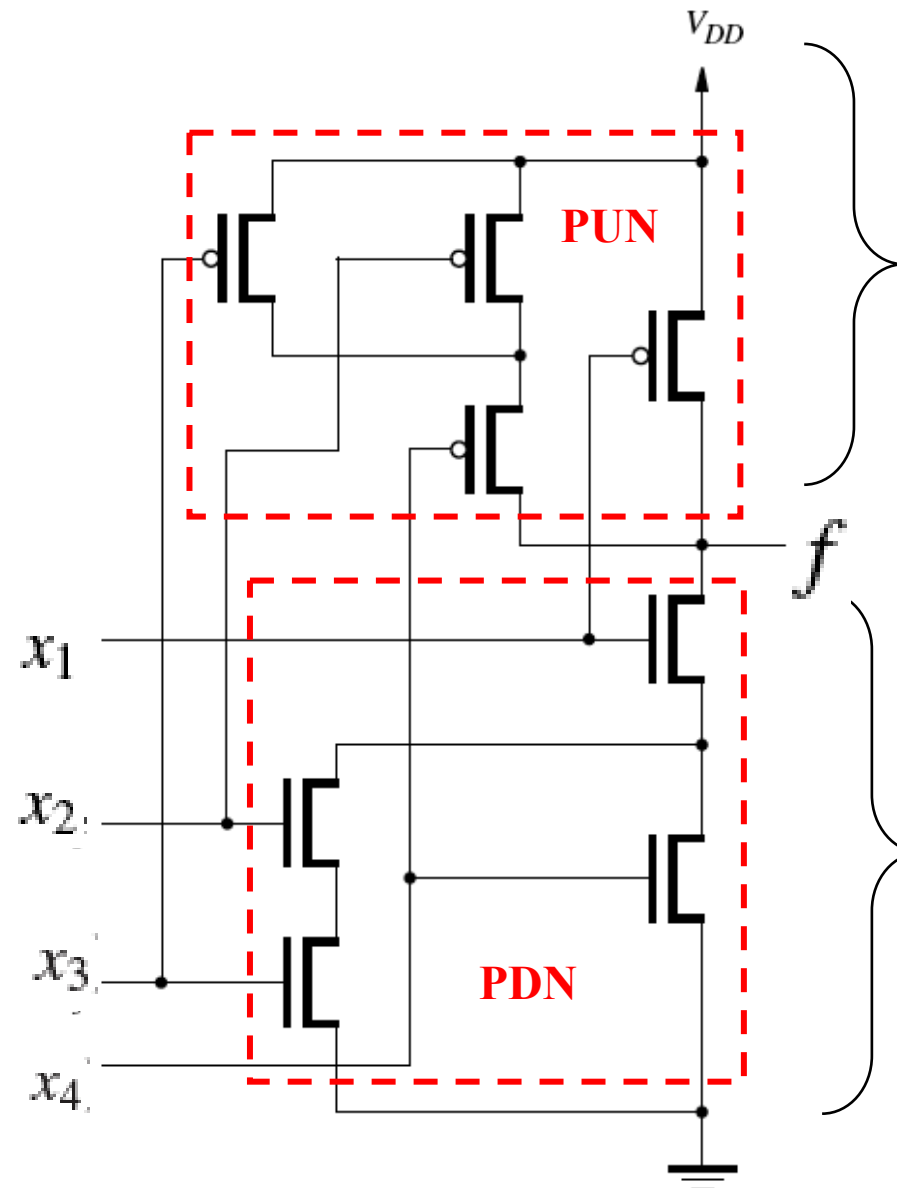
x_1 in series with (x_2 and x_3 in parallel)

Example 2

Implement the below function using CMOS logic

$$f = \bar{x}_1 + (\bar{x}_2 + \bar{x}_3)\bar{x}_4$$

Example 2



Implement

$$f = \bar{x}_1 + (\bar{x}_2 + \bar{x}_3)\bar{x}_4$$

**All the inputs are already in complement
 x_1' in parallel with [x_2' and x_3' in parallel]
in series with x_4']**

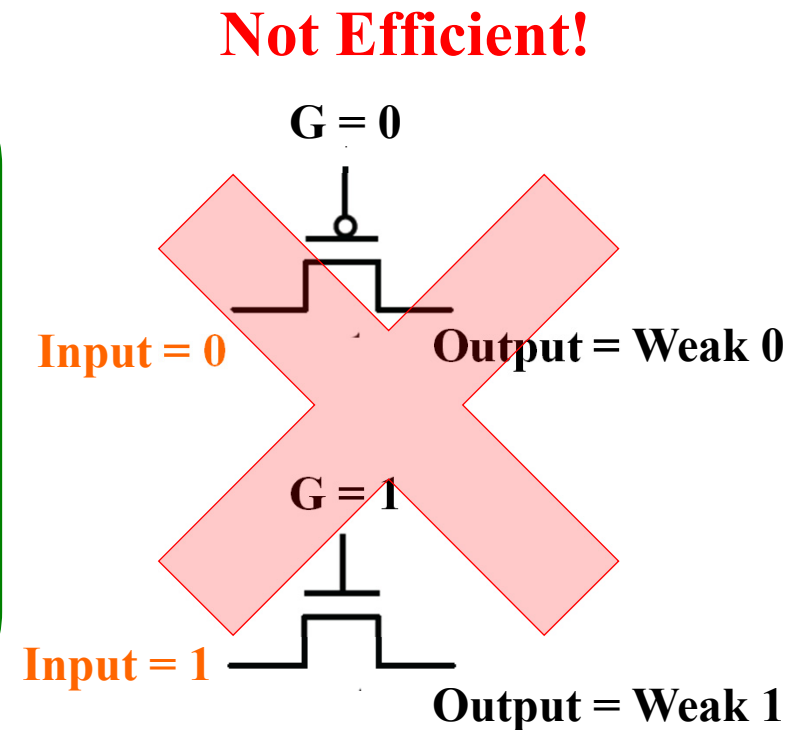
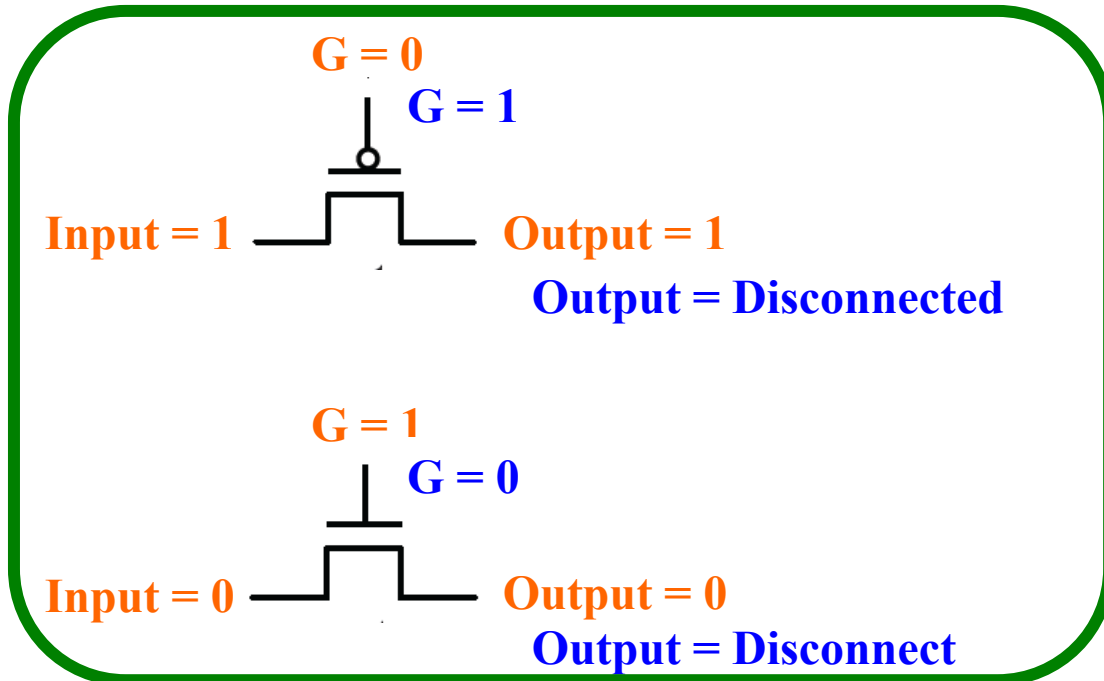
Implement

$$\bar{f} = x_1(x_2x_3 + x_4)$$

**x_1 in series with [x_2 and x_3 in series] in
parallel with x_4]**

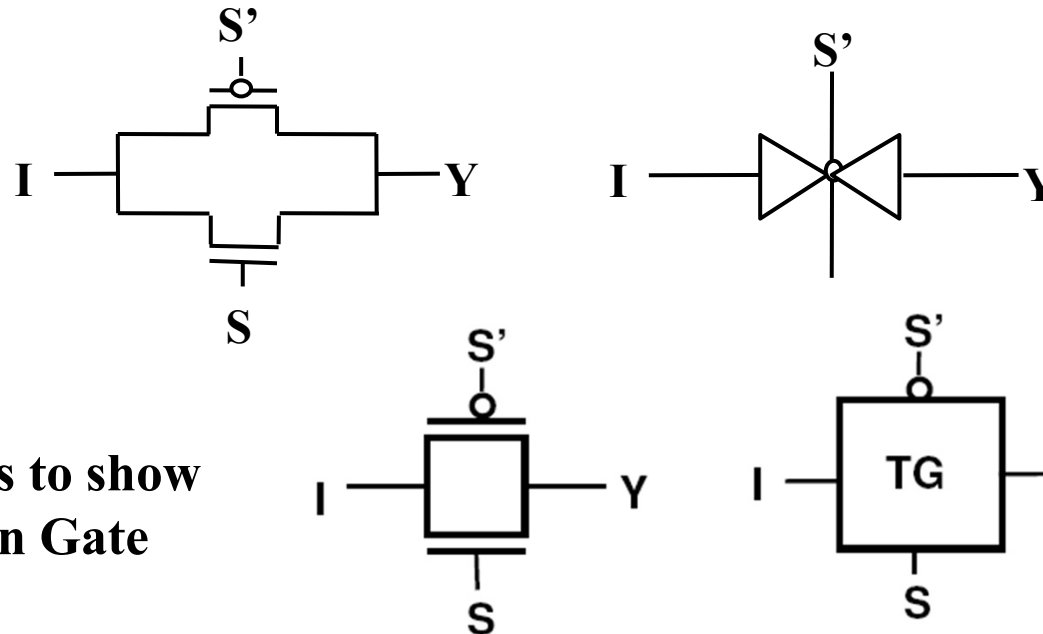
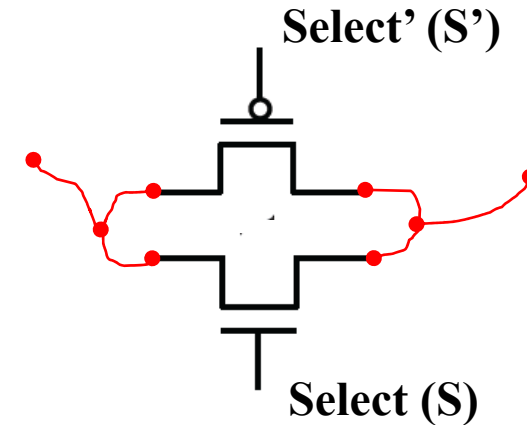
Another way: Pass Transistor Logic

- Use PMOS or NMOS transistors to pass bits (1 or 0) from input to output
- Control the gate (G) input to pass or not to pass the bit
- Either Source (S) or Drain (D) can be used for input or output (bi-directional)
- Notice that there is no supply (V_{DD}) or GND connection at all
- NMOS works well connecting “logic 0”
- PMOS works well connecting “logic 1”



Transmission Gate (is also a switch)

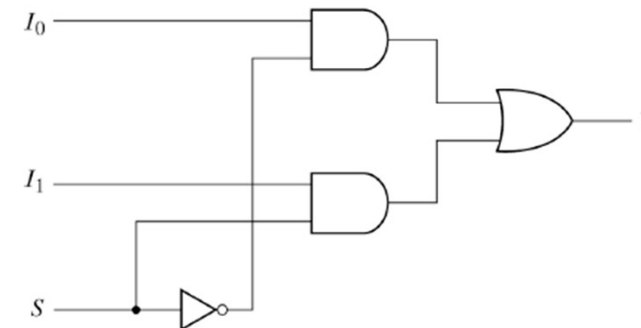
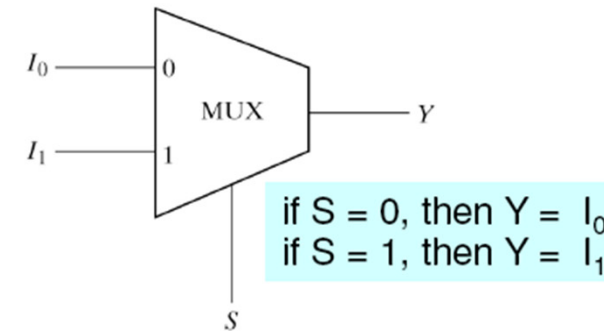
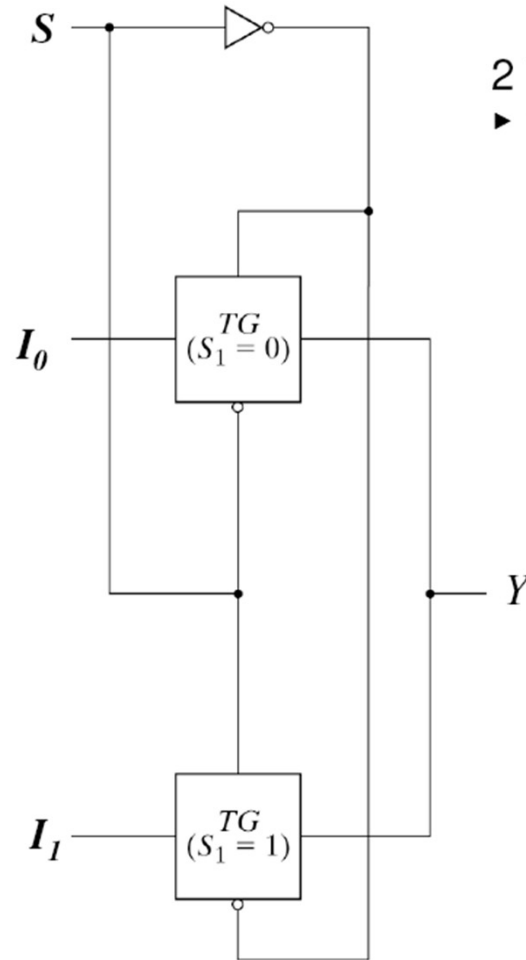
- To pass both 0 and 1 efficiently
 - Connect one NMOS and one PMOS like this →
 - Label G (gate) inputs as complementary Select (S) inputs
 - NMOS switches work well only when connecting to 0
 - PMOS switches work well only when connecting to 1



Other symbols to show
a Transmission Gate



- 2-to-1 multiplexer



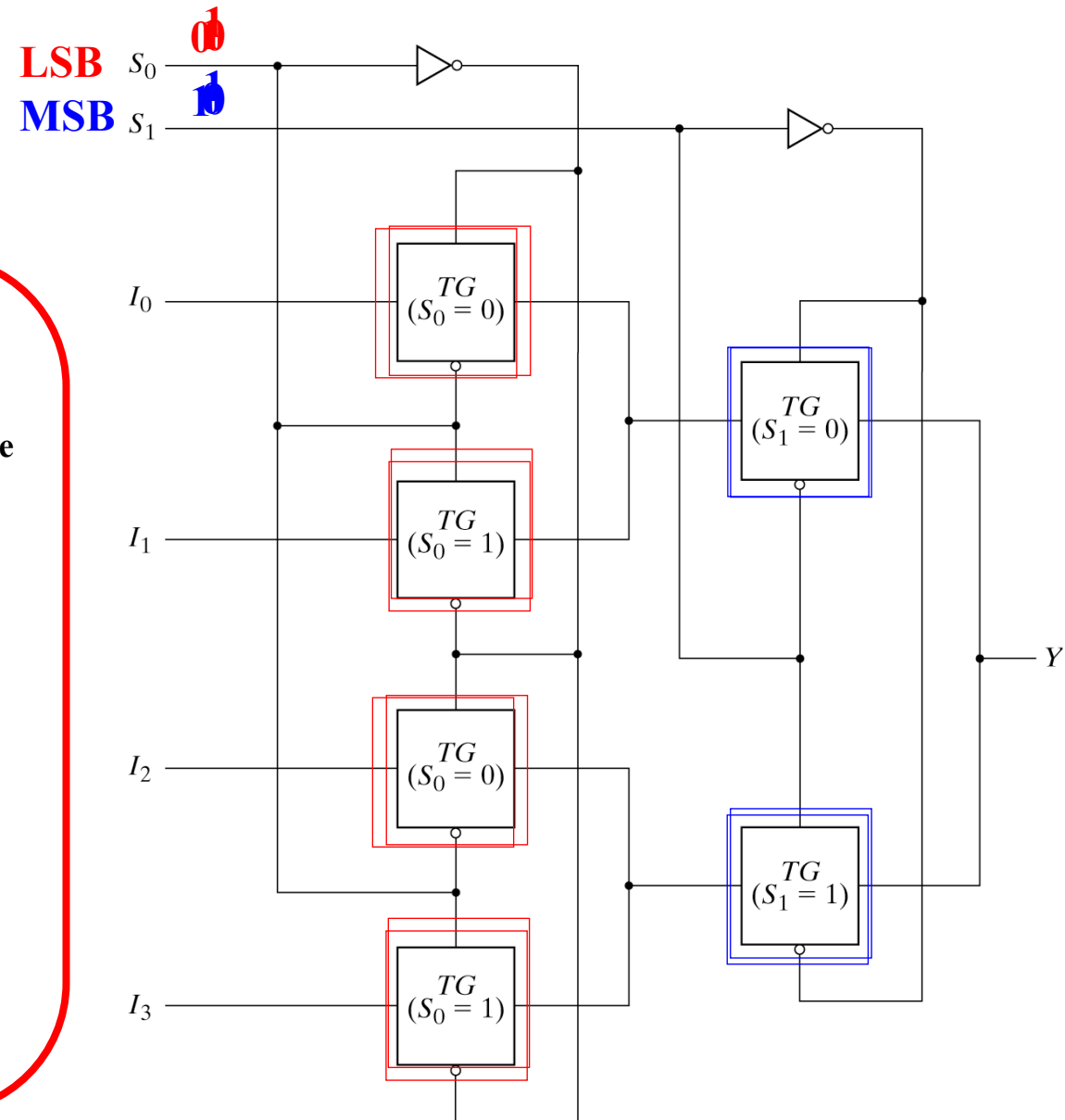
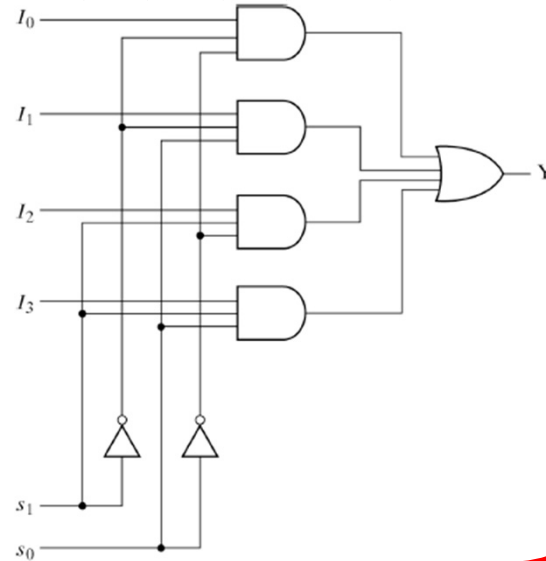
Multiplexer by CMOS Transmission gate (TG)



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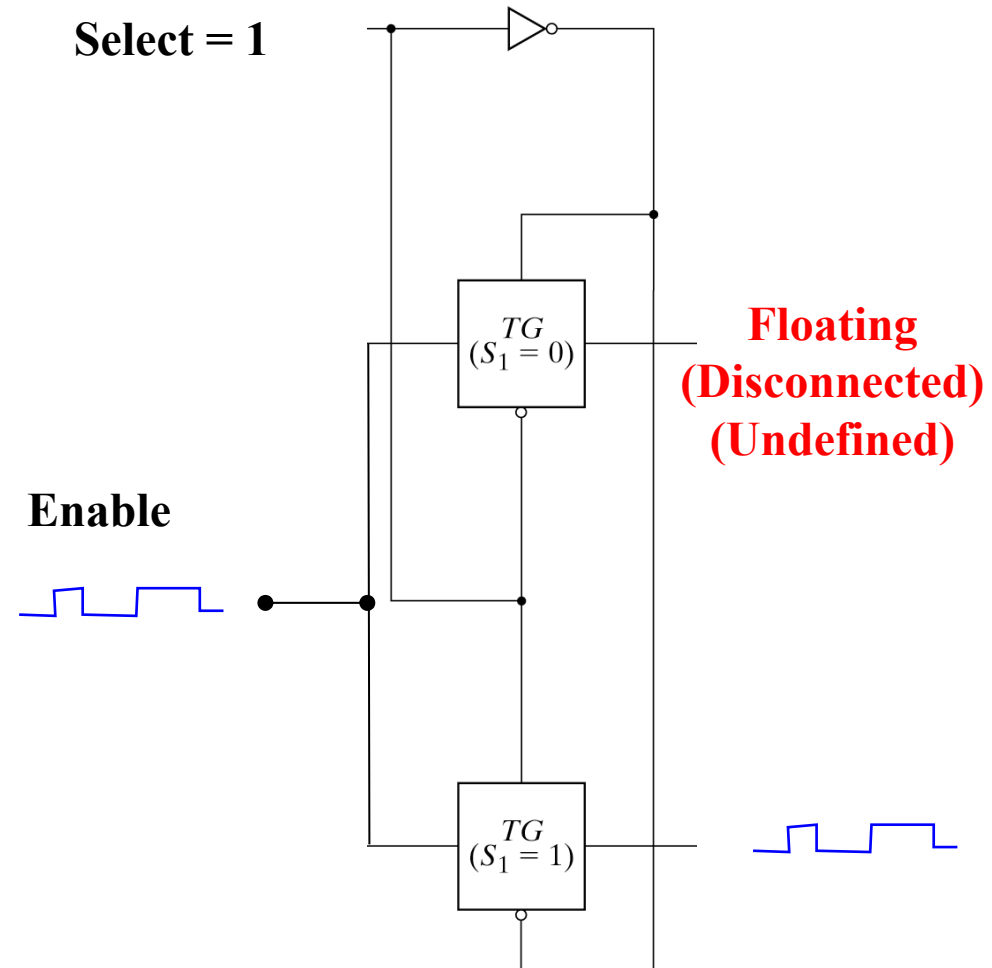
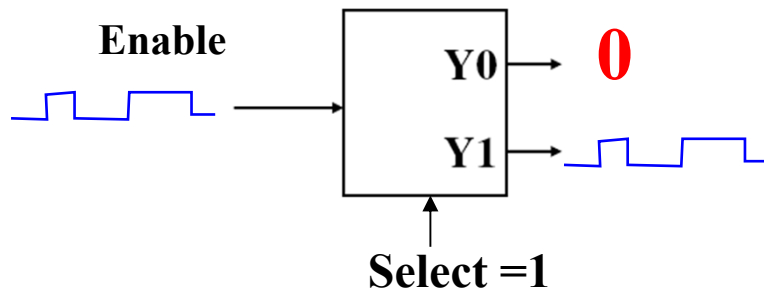
- 4-1 multiplexer
 - (2+4) TG and 2 inverters
 - » 4 equivalent gates
 - by NAND-NAND gate
 - » 9 equivalent gates

3 input NAND/NOR gate = 1.5 Equivalent gate
4 input NAND/NOR gate = 2 Equivalent gate
Each inverter = 0.5 Equivalent gate
Total = $(4 \times 1.5) + (1 \times 2) + (2 \times 0.5) = 9$



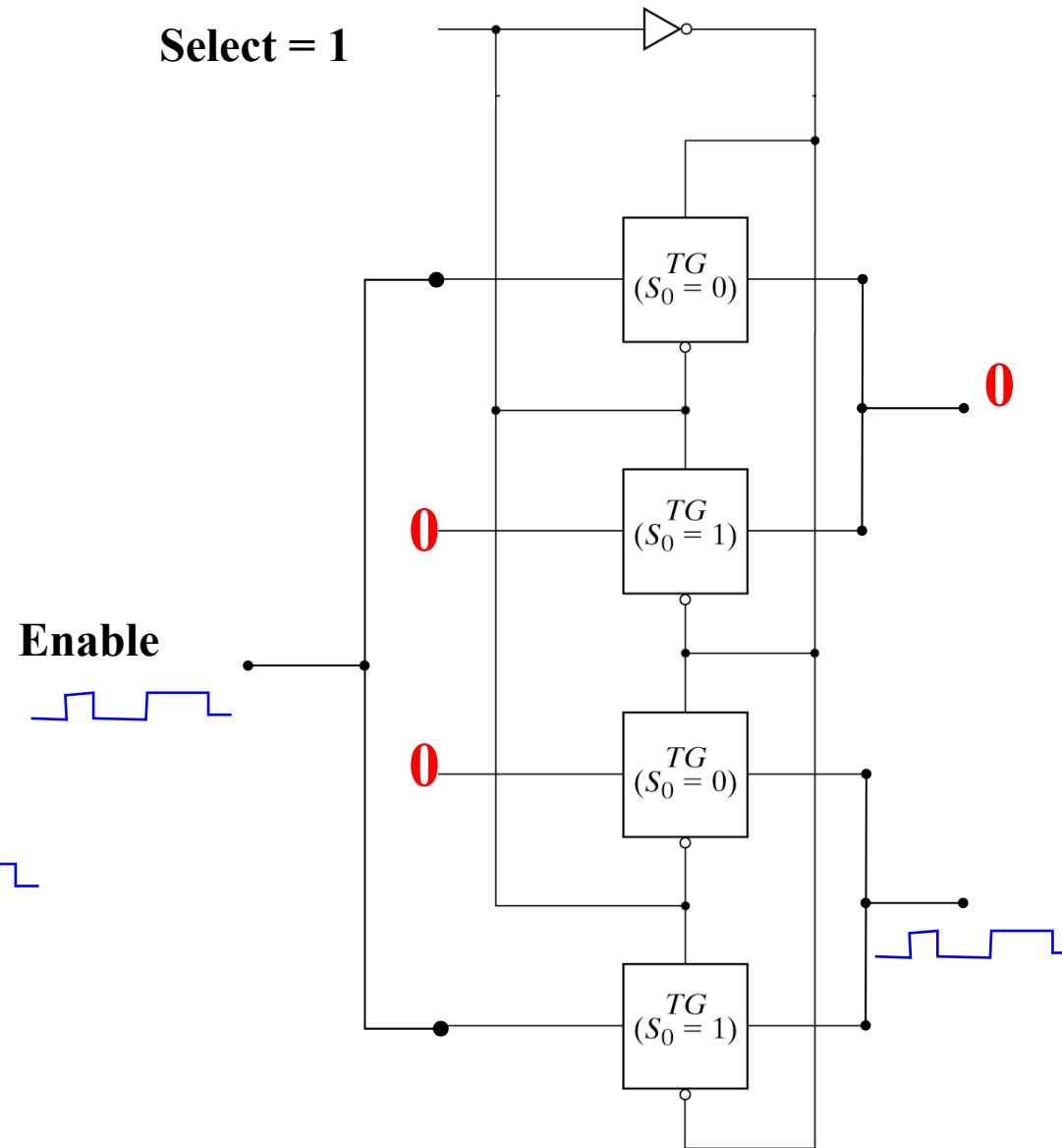
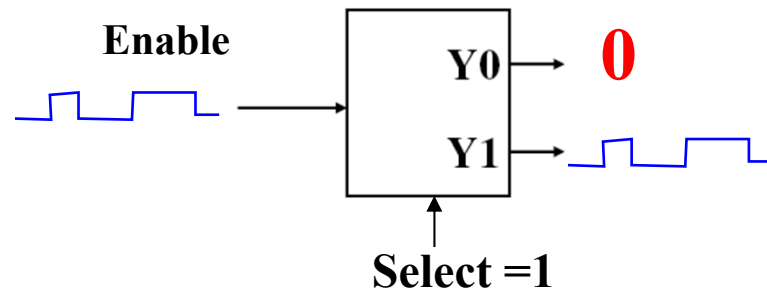
Demultiplexer by Transmission gates

- 1-2 demultiplexer
 - 1 select input
 - 1 enable input
 - 2 outputs
 - » Y goes to 1 output
 - » the other is floated

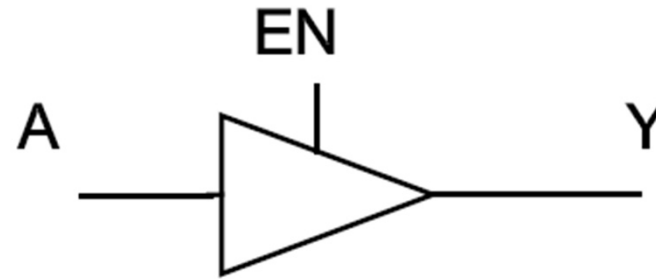


Demultiplexer by Transmission gates

- **1-2 demultiplexer**
 - **1 select input**
 - **1 enable input Y**
 - » 1 logic 0 input
 - **2 outputs**
 - » Y goes to 1 output
 - » 0 goes to the other
 - **4 TGs and 1 NOT**
 - » 2.5 equivalent gates

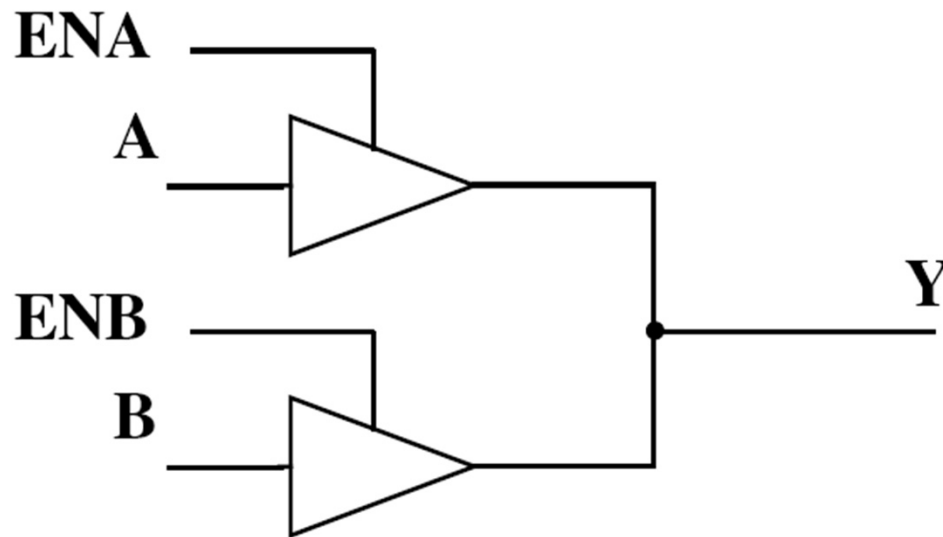


- When $EN=1$, then $Y=A$
- When $EN=0$, then Y is disconnected from A and Y is *undriven*, we call it the high-impedance state (State 'Z'), so $Y='Z'$, at high impedance state.
- Tri-state means now Y can be at 3 different states: 0, 1, 'Z'



Using Tri-state gates

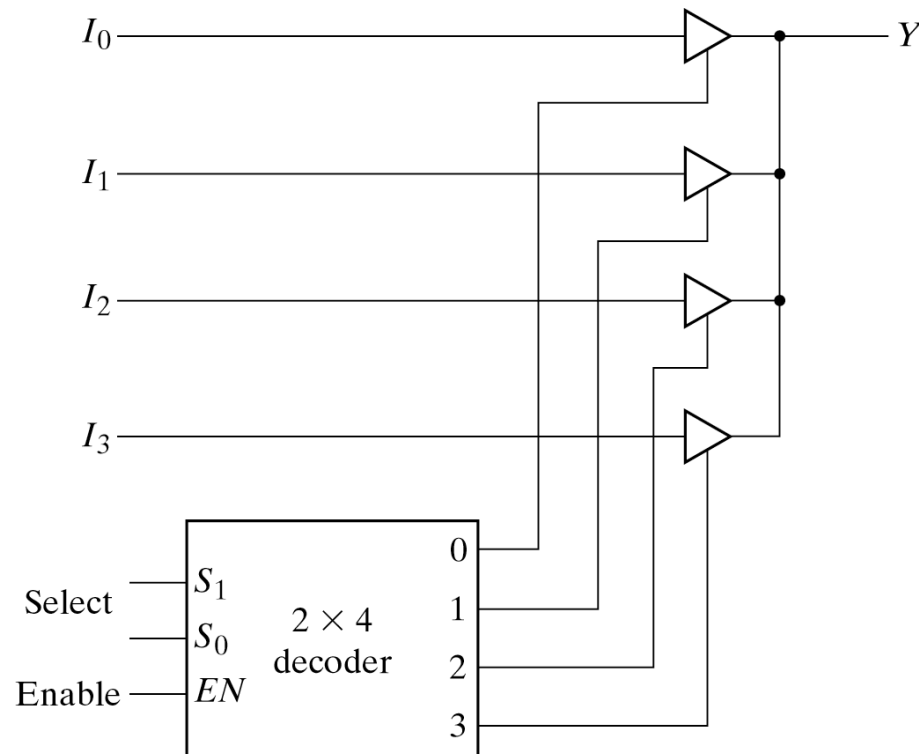
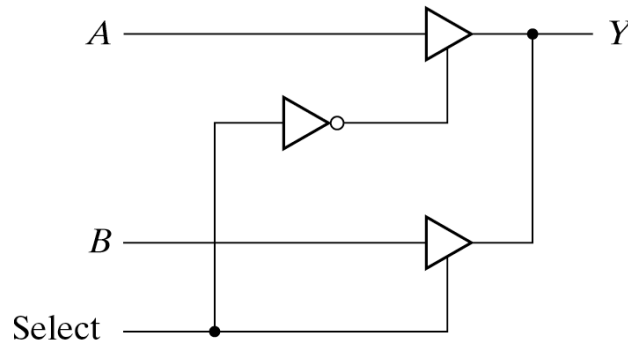
- The output of tri-state gates can be tied together.
 - As long as only one output has value (0 or 1) and the other outputs are at high impedance states ('Z')
- Must make sure that ENA and ENB are not both '1' at the same time, or Y will be driven from multiple active sources



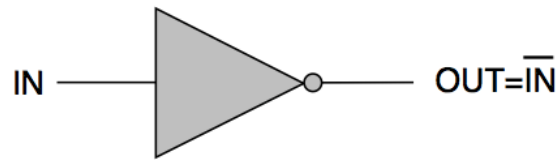
ENB	ENA	Y
0	0	Z (high impedance)
0	1	A
1	0	B
1	1	X (very bad idea!!!!)

Multiplexer by Tri-state Gates

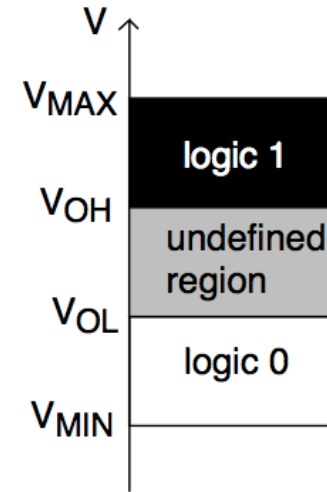
- **2-to-1 multiplexer**
 - **2 tri-state gates**
 - » only one is enabled
- **4-to-1 multiplexer**
 - **4 tri-state gates**



In digital circuits, digitally-encoded information is represented by means of two distinct voltage ranges:

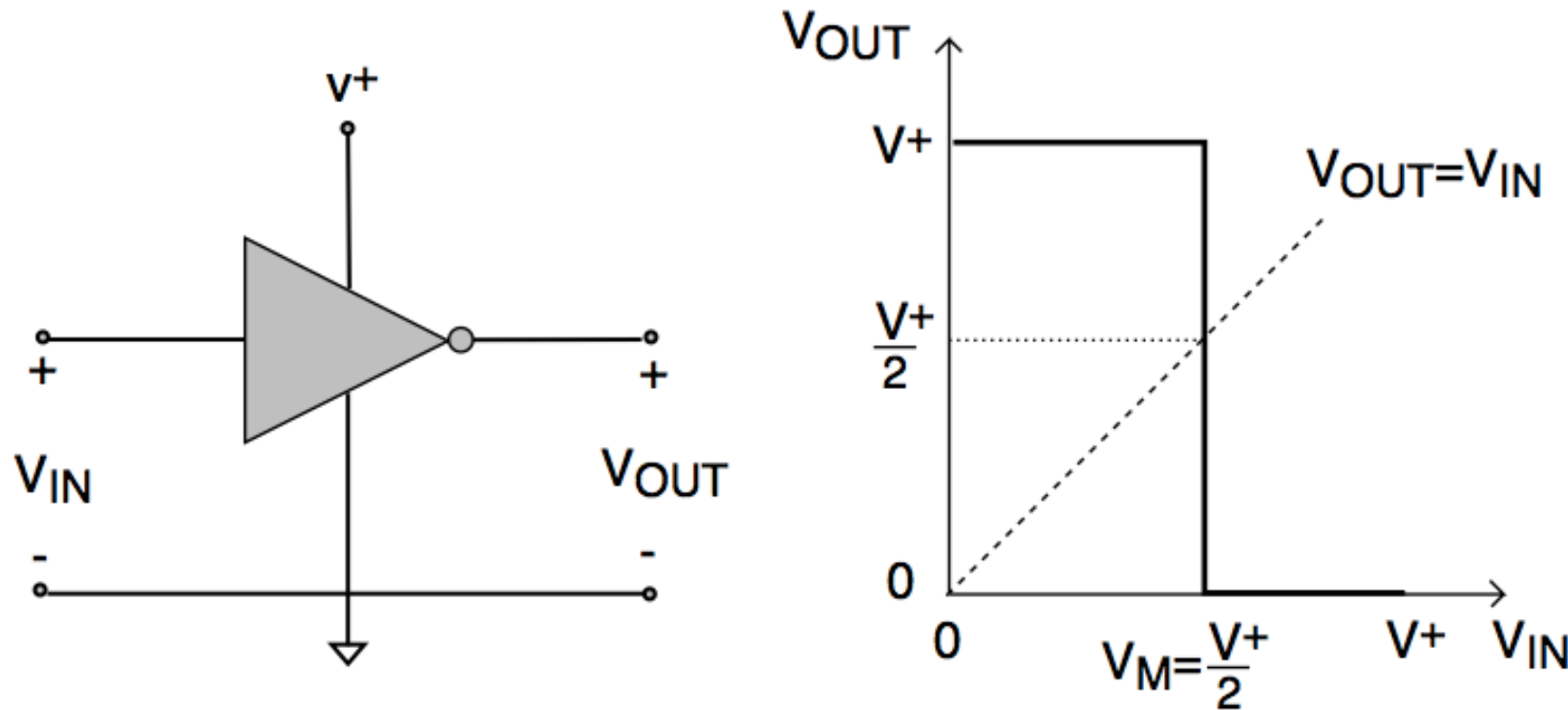


IN	OUT
0	1
1	0



- **Logic 0:** $V_{\text{MIN}} \leq V \leq V_{\text{OL}}$
- **Logic 1:** $V_{\text{OH}} \leq V \leq V_{\text{MAX}}$
- **Undefined logic value:** $V_{\text{OL}} \leq V \leq V_{\text{OH}}$

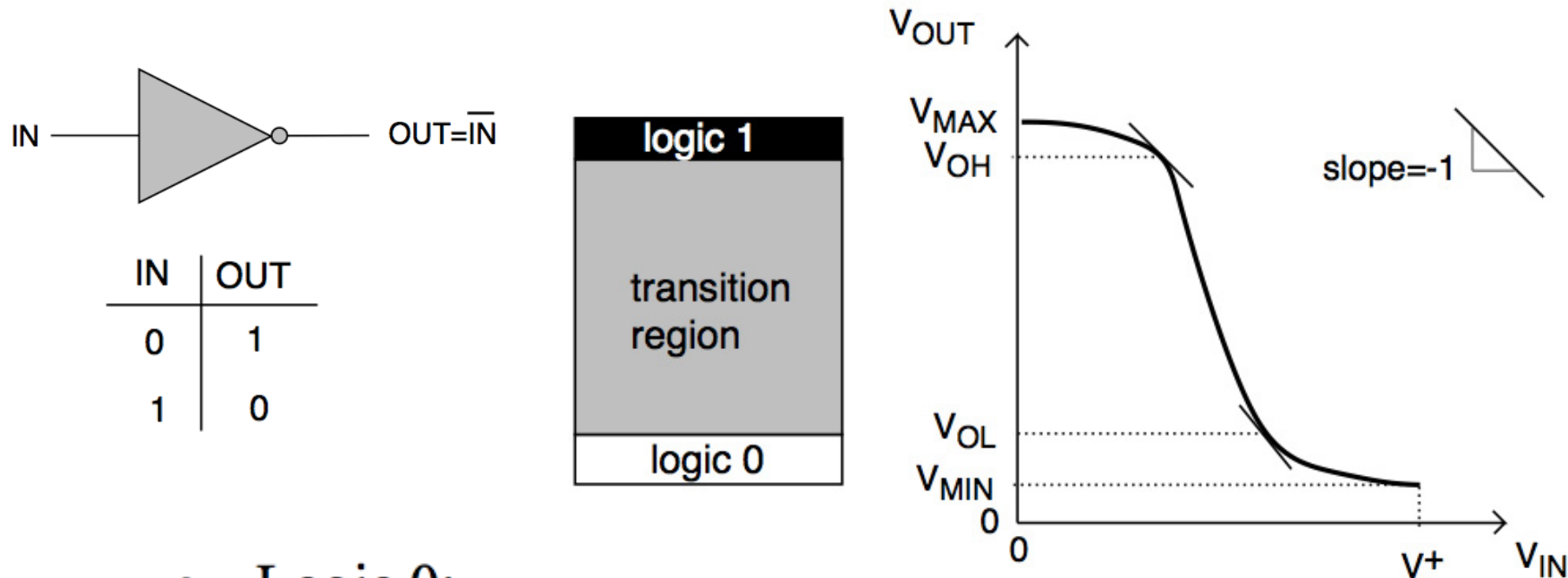
Input-Output relation (Ideal Inverter)



Define *switching point* or *logic threshold* :

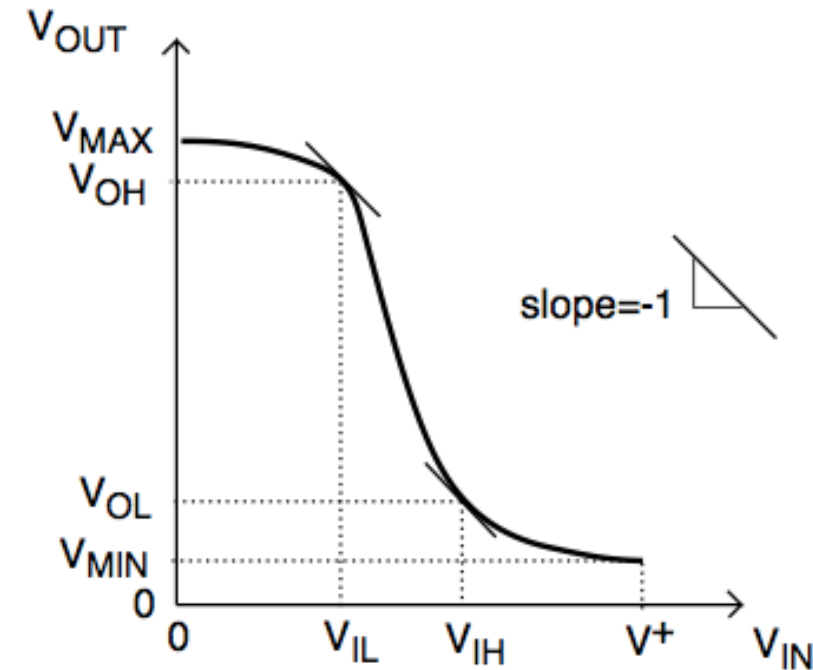
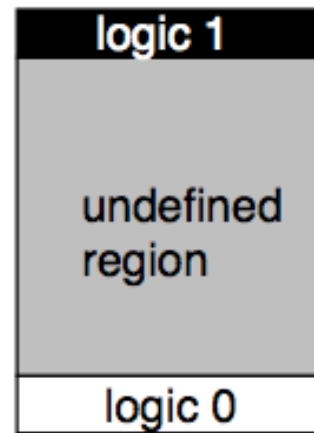
- $V_M \equiv$ input voltage for which $V_{OUT} = V_{IN}$
 - For $0 \leq V_{IN} < V_M \Rightarrow V_{OUT} = V^+$
 - For $V_M < V_{IN} \leq V^+ \Rightarrow V_{OUT} = 0$

Input-Output relation (Real Inverter)



- Logic 0:
 - $V_{MIN} \equiv$ output voltage for which $V_{IN} = V^+$
 - $V_{OL} \equiv$ smallest output voltage where slope = -1
- Logic 1:
 - $V_{OH} \equiv$ largest output voltage where slope = -1
 - $V_{MAX} \equiv$ output voltage for which $V_{IN} = 0$

Two other important voltages:

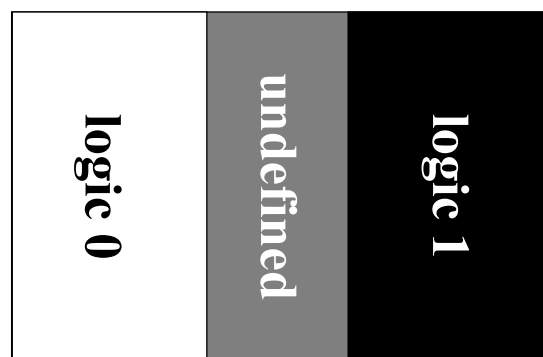
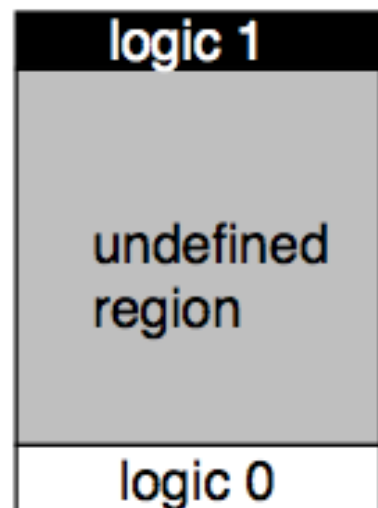
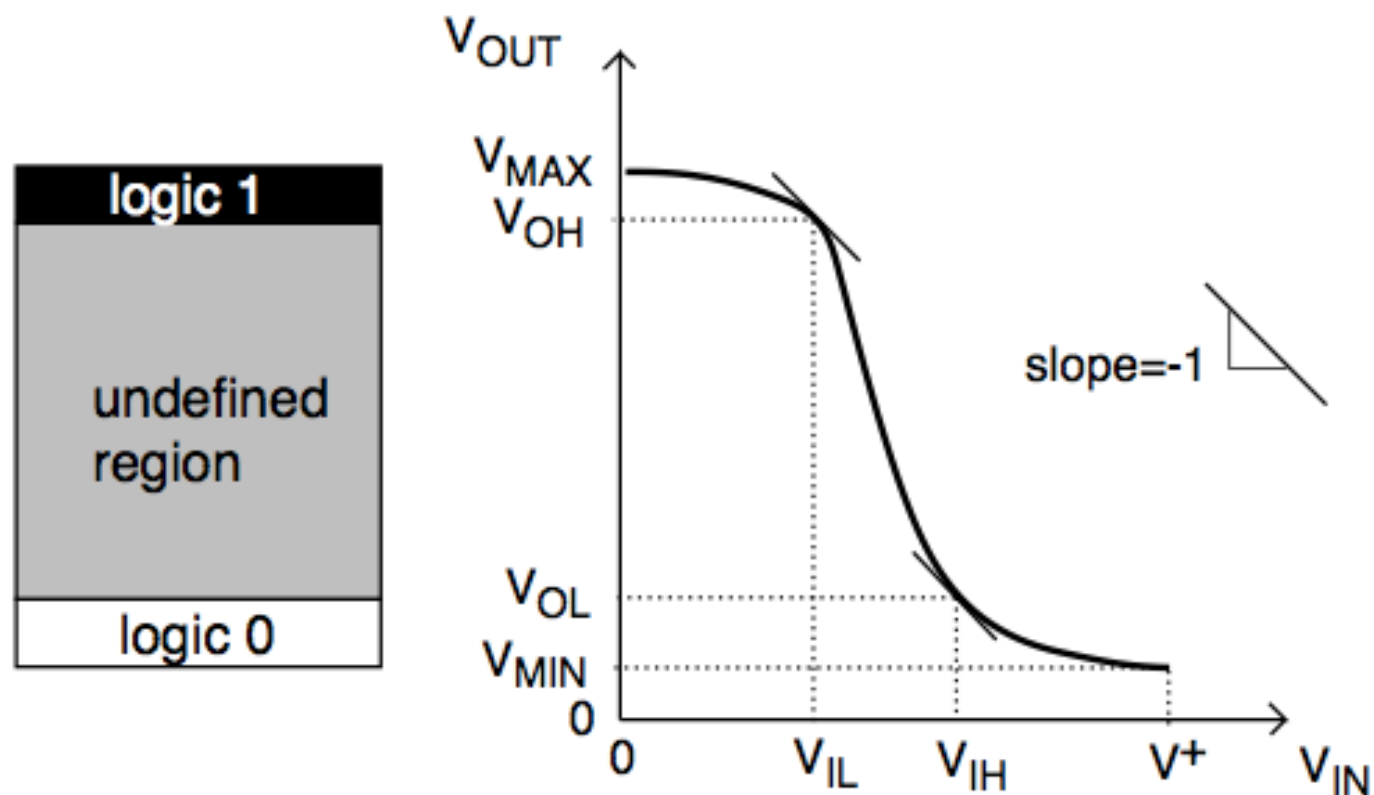


range of input values
that produce valid logic 1

range of input values
that produce valid logic 0

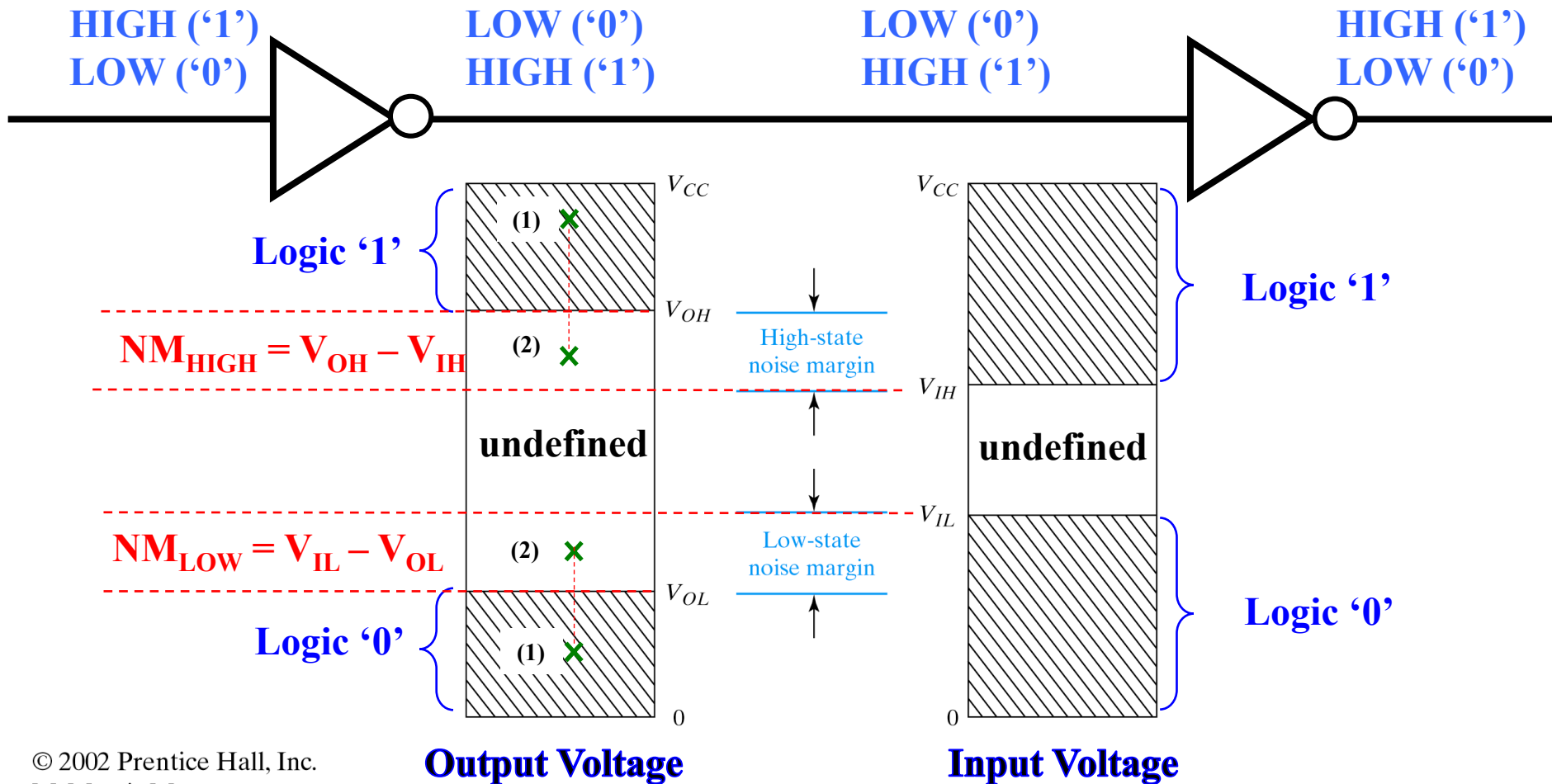
$V_{IL} \equiv$ smallest input voltage where slope = -1

$V_{IH} \equiv$ highest input voltage where slope = -1

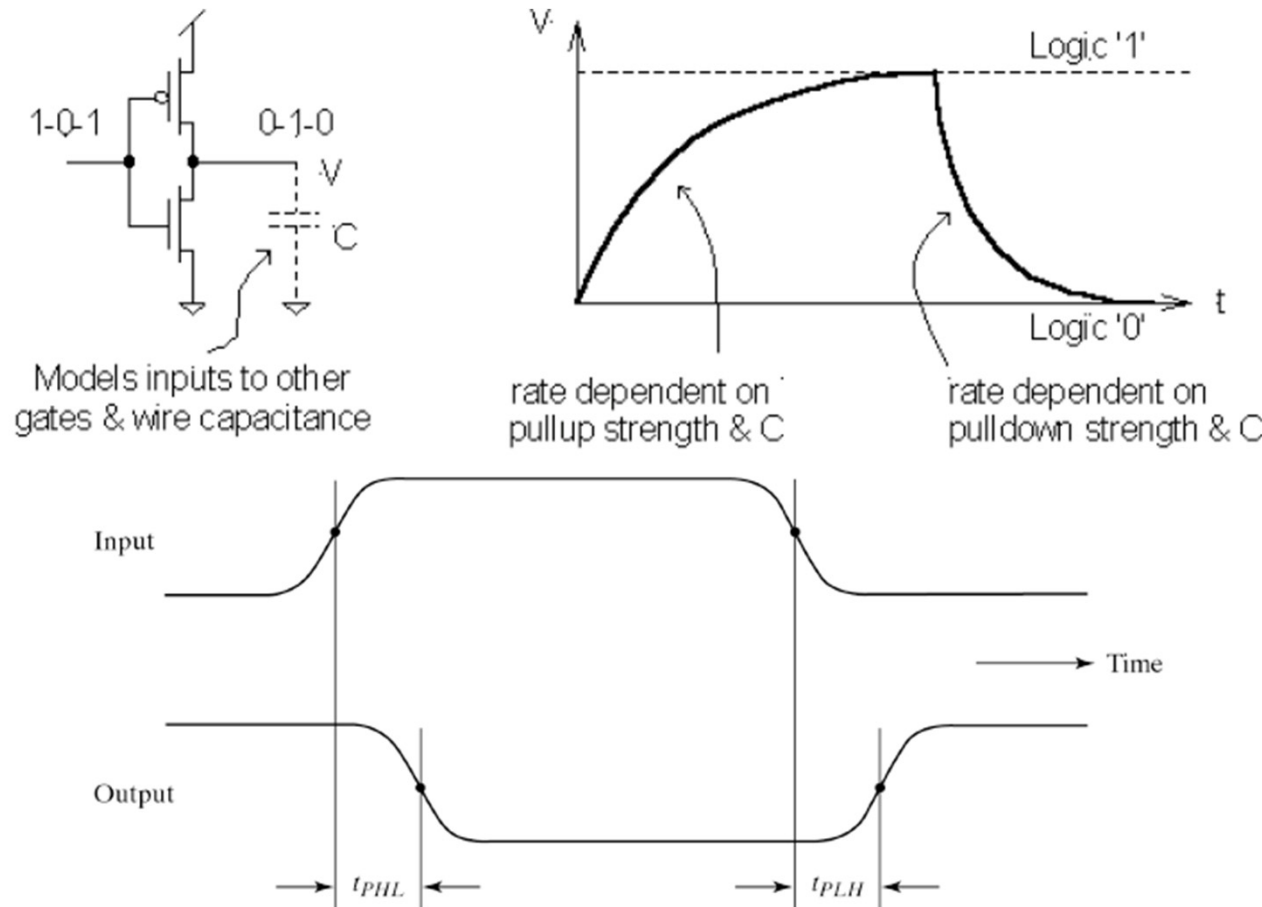


Noise Margin (NM)

Noise is undesired and causes output voltage (1) to change its value (diagram).
As long as the new value (2) is within the noise margin, result will be correct!



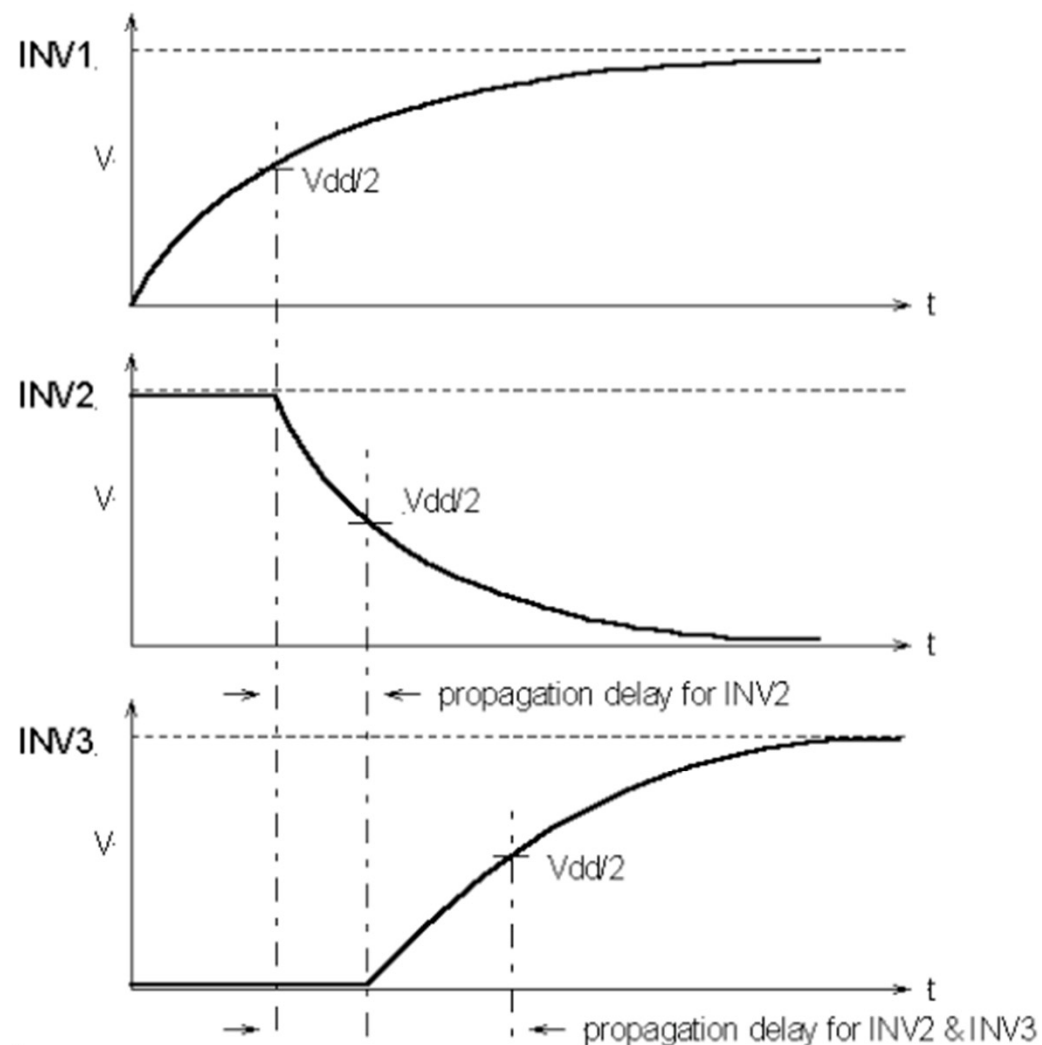
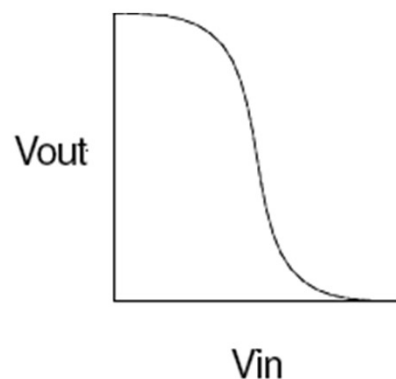
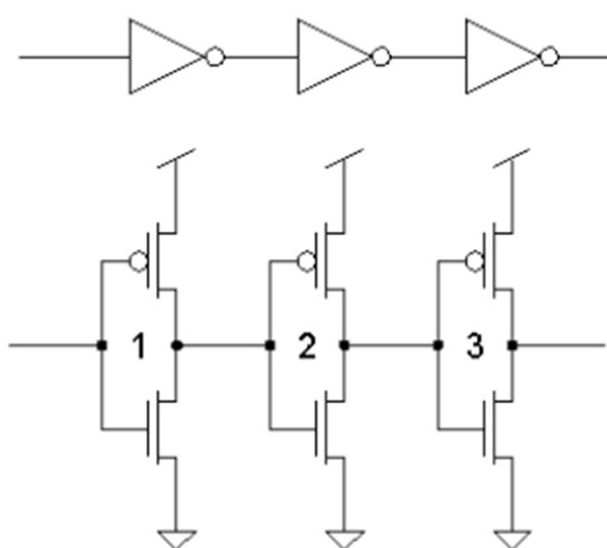
- **Propagation delay: time for signal from input to output**
 - t_{PHL} : for output to be changed from High to Low
 - t_{PLH} : for output to be changed from Low to High



Propagation Delay



- **Cascaded gates:**



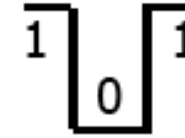
In general:
prop. delay = sum of individual prop. delays of gates in series.

- **Hazards/glitches: unwanted switching at the outputs**
 - **Occur when different paths through circuit have different propagation delays**
 - **Dangerous if logic causes an action while output is unstable**
 - **May need to guarantee absence of glitches**
- **Usual solutions**
 1. **Wait until signals are stable (by using a clock): preferable (easiest to design when there is a clock – synchronous design)**
 2. **Design hazard-free circuits: sometimes necessary (clock not used – asynchronous design)**

Types of Hazards

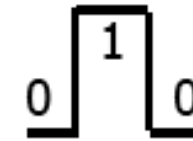
- Static 1-hazard

- Input change causes output to go from 1 to 0 to 1



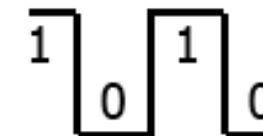
- Static 0-hazard

- Input change causes output to go from 0 to 1 to 0



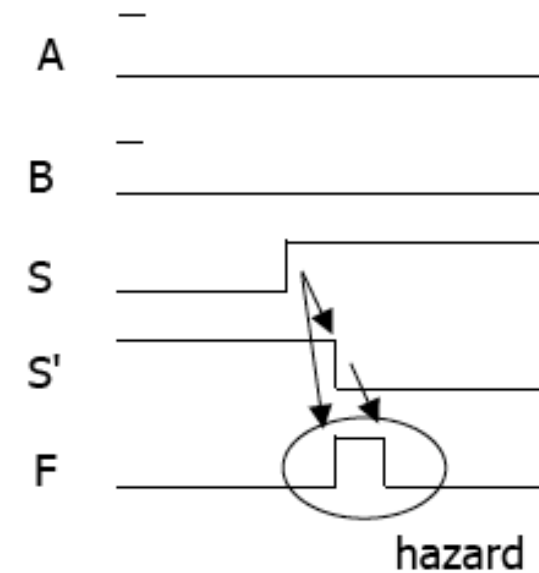
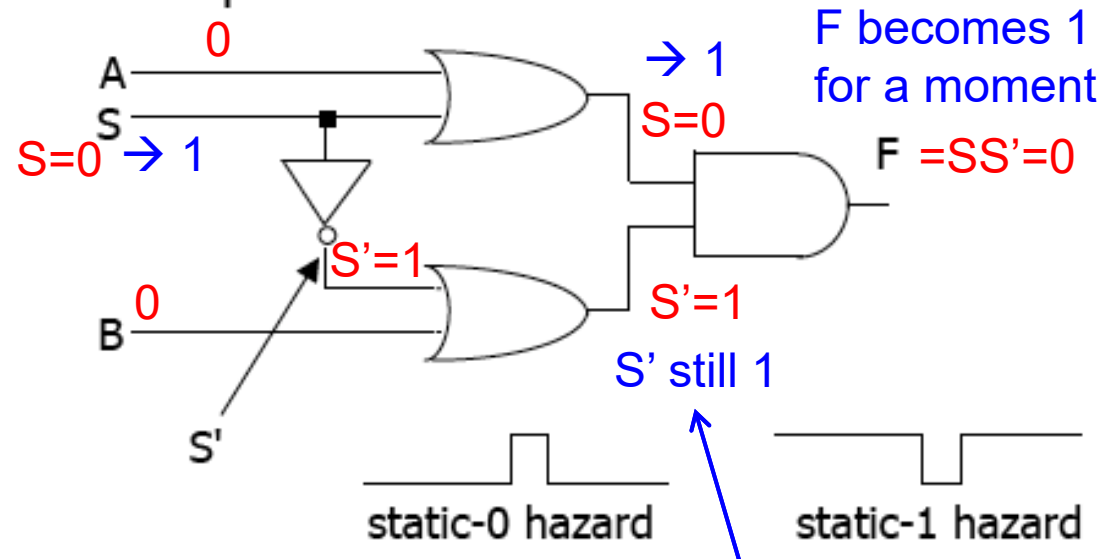
- Dynamic hazards

- Input change causes a double change
from 0 to 1 to 0 to 1 OR from 1 to 0 to 1 to 0



Static Hazards

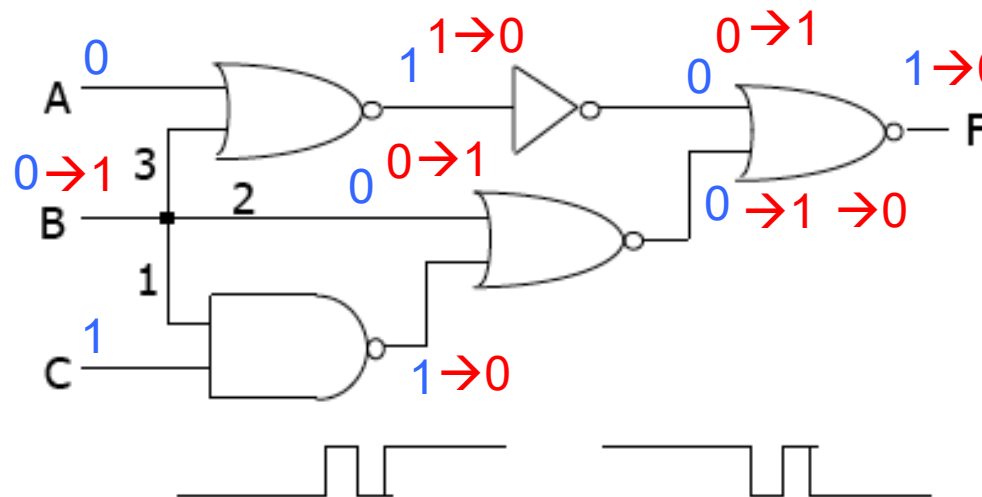
- Due to a literal and its complement momentarily taking on the same value
 - Thru different paths with different delays and reconverging
- May cause an output that should have stayed at the same value to momentarily take on the wrong value
- Example: When S changes from 0 to 1



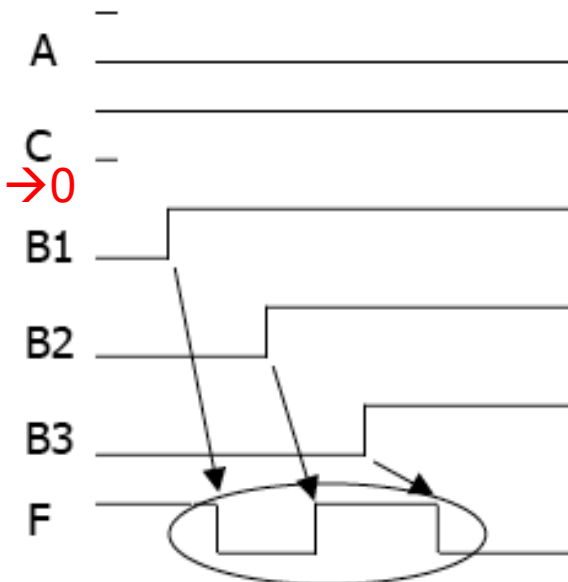
New value $S' \rightarrow 0$ has not reached there yet because the inverter delay!

Dynamic Hazards

- Due to the same versions of a literal taking on opposite values
 - Thru different paths with different delays and reconverging
- May cause an output that was to change value to change 3 times instead of once
- Example: **When B changes from 0 to 1**



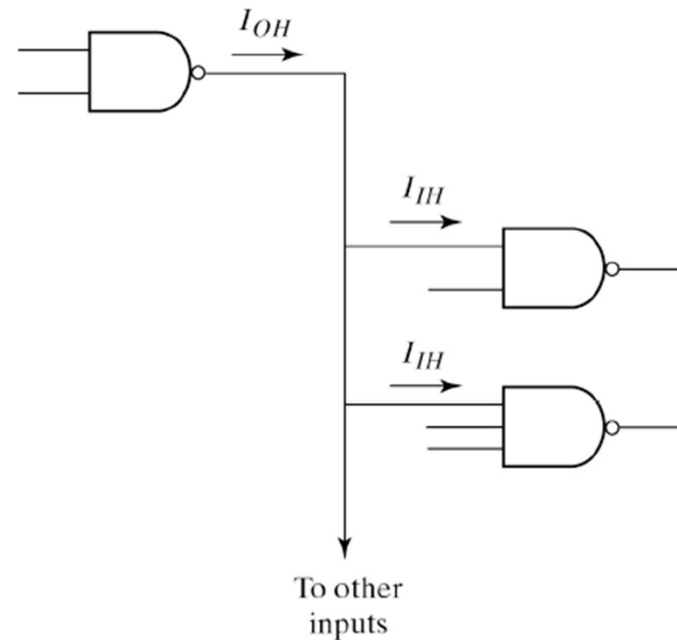
dynamic hazards



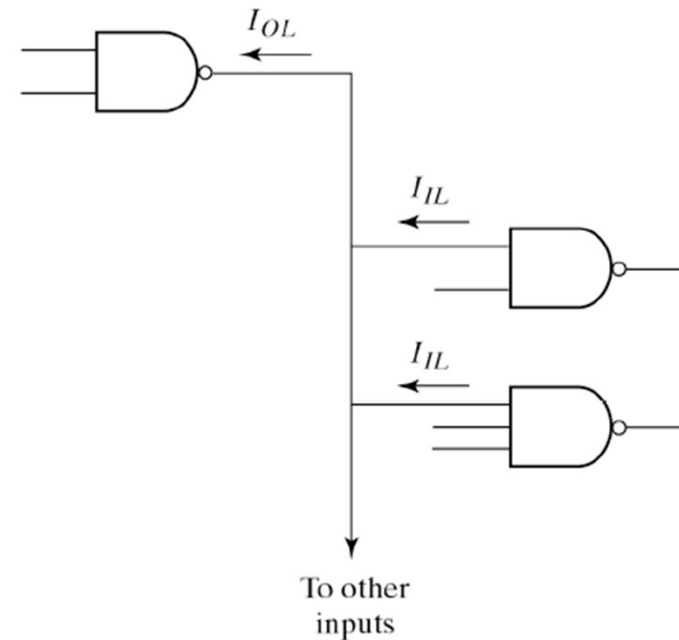
hazard



- How many gates can be driven by one gate...
- – $I_{OH} = 400 \mu\text{A}$, $I_{IH} = 40 \mu\text{A}$, Fan-out = $I_{OH}/I_{IH} = 10$
- – $I_{OL} = 16 \text{ mA}$, $I_{IL} = 1.6 \text{ mA}$, Fan-out = $I_{OL}/I_{IL} = 10$



(a) High-level output



(b) Low-level output



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Thank you for listening !

